MED-ALALAY: ANDROID-BASED MEDICATION REMINDER APPLICATION INTEGRATING GIS-BASED MAPPING FOR ELDERLY INDIVIDUALS

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In Partial Fulfillment of the Requirements for the Degree of
Bachelor of Science in Information Technology
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Application Development

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M.E.A

A.C.D

G.B.T

ABSTRACT

Medication non-adherence among elderly individuals remains a major global and local healthcare challenge, contributing to increased hospitalization, complications, and mortality. This study developed MED-ALALAY, an Android-based medication reminder application that integrates voice notifications and GIS-based mapping to support elderly individuals in managing their prescribed medication schedules. The system was developed using the Agile Scrum methodology with React Native and TypeScript, featuring voice-based medication reminders, caregiver or dependent monitoring, health note management, appointment scheduling, and navigation to nearby pharmacies, clinics, and hospitals. The system was evaluated using the Technology Acceptance Model (TAM) with 50 end-users and the ISO 25010 Software Quality Standard with 3 IT professionals. Results showed high user acceptability (overall TAM mean = 3.19), with strong user attitudes toward the Application's usefulness and simplicity. ISO 25010 evaluation yielded a composite mean score of 3.74, indicating excellent functional suitability, performance efficiency, usability, and portability. Findings demonstrate that voice notifications improved medication adherence, while GIS integration enhanced accessibility to healthcare services. Overall, MED-ALALAY proved effective as an accessible, reliable, and user-friendly mobile health solution for elderly individuals to manage complex medication regimens.

Keywords: *Medication Adherence, Elderly Care, Voice Notification, GIS-Mapping, Mobile, Health Application, Android Development*

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DEFINITION OF TERMS

Terms	Definition
Android-Based Application	A mobile application that works on devices with Android operating systems and is accessible with a wide range of smartphones and tablets.
Application Programming Interface (API)	A set of protocols and tools that enables different software applications to communicate and exchange data with each other.
Data Analytics	A system function that tracks, analyzes, and visualizes drug adherence patterns, such as doses taken, missed, and inventory levels.
Database	A system function that tracks, analyzes, and visualizes drug adherence patterns, such as doses taken, missed, and inventory levels.
Dependent	A family member, caregiver, or authorized individual who observes and assists the elderly individual's

	medication adherence using the application.
GIS (Geographic Information System)	A system for collecting, storing, processing, analyzing, managing, and presenting geographical or geographic data, which was utilized in this study to locate nearby healthcare facilities.
GPS (Global Positioning System)	A satellite-based navigation system used to determine the real-time geographic location of users for mapping and navigation purposes.
ISO 25010	An international standard for software product quality that defines characteristics of quality such as functional suitability, performance efficiency, usability, reliability, security, maintainability, and portability.
Medication Adherence	The period that patients take medicines as prescribed by medical professionals, including the correct quantity, time, and frequency.
Medication Non-adherence	The failure to follow prescribed medication treatments, which can lead

	to serious health problems, increased hospitalizations, and higher mortality rates.
Medication Regimen	A prescribed schedule that specifies which medicines to take, in what quantities, when, and for how long.
Medication Reminder System	A digital system that notifies users about their prescribed medication schedules through visual alerts, notifications, or voice prompts to improve medication adherence.
Mobile Health (mHealth)	The use of mobile devices and applications to support healthcare services, health monitoring, and the delivery of medical information.
Reminder Notification	A system notice that encourages users to take their prescribed medicines at specific periods using both visual and audio approaches.
Text-to-Speech (TTS) Technology	A type of assistive technology that reads digital texts while converting readable drug information into audible voice notifications.

User Interface (UI)	The visual layout and interactive components of the application that allow users to navigate and interact with the system easily and efficiently.
Voice Notification	A type of assistive technology that reads digital texts while converting readable drug information into audible voice notifications.

CHAPTER I

INTRODUCTION

Project Context

Medicine is an essential component of health, as it is a primary tool for managing, preventing, and treating illnesses. Positive health outcomes are achievable if medication is taken as prescribed by the doctor. The effectiveness of drugs depends on how they are taken correctly, which is why medication adherence is important for improving quality of life and life expectancy. Non-adherence persists as a major problem in the world today and has been associated with higher mortality, worsening of medical conditions, and higher hospitalization rates, especially among elderly individuals.

One of the essential parts in achieving positive health outcomes is how well individuals take their medications as prescribed by their doctors. It is not only effective with the medicine formula but also if it is managed correctly by the prescribed schedule. Elderly individuals are the ones who have the most challenges because they usually deal with a variety of medical problems that require complicated drug regimens. It might be difficult for older persons to take their medications on their own due to cognitive decline, vision impairment, physical vulnerability, and neglect. In the Philippines, many elderly individuals rely on their memory. This highlights the importance of having a reliable and effective way to help elderly individuals adhere to their prescribed schedules. It is more possible that elders may forget or manage their dosages incorrectly. Globally, the aging population has led to a significant demographic shift with broad impacts. As populations age, birth rates decrease, and life expectancy increases, leading to an increasing number of elderly individuals (Ismail et al.,

2021). Among the many difficulties arising from demographic shifts are challenges in healthcare systems and the economy (National Institute on Aging, 2007). By addressing the approach, innovative solutions that support inclusive social policies, intergenerational solidarity, and healthy aging might be developed (Mohd & Haque, 2024).

Republic Act No. 7432 defines "senior citizens" as any resident citizen who has reached at least 60 years of age, a term used to describe older persons in the Philippines. Despite less than ten percent of the population being elderly, the Philippines can be classified as a young society in terms of age distribution. According to the Philippine Statistics Authority's 2015 census, an estimated 7.5 million individuals are elderly, accounting for about 7.5 percent of the nation's total population. There is a trend of better functional health and increased life expectancy among Filipino elderly people, despite reports of significant discrepancies in their health status (Cruz & Fabon, 2019).

Positive feelings fade as people age, as their health inhibits development and can be challenging. In addition, many studies have found a modest decline in life satisfaction among the elderly, consistent with the association between age and life satisfaction (Galiana et al., 2016). The World Health Organization (2003) reported that almost half of all patients with chronic illnesses fail to follow their prescribed medication.

According to (Gellad et al., 2011), older adults face difficulties in following their prescribed medication schedules due to complex treatment regimens or cognitive impairment. In the elderly, medicines are commonly prescribed to enhance quality of life, increase lifespan, and treat or mitigate diseases. The elderly's

failure to follow their prescriptions often results in negative clinical and economic outcomes (Bal Krishnan, 1998).

Medication adherence is a serious health matter that needs to be addressed, especially for elderly individuals who have a higher rate of taking medicine for maintenance or taking prescribed medication. The studies show that it is hard for elders to take their medication on a prescribed schedule, particularly those who experience forgetfulness or who live independently due to a lack of support or assistance from their family, since family members, like their offspring, have their own lives or are out for something.

Individuals, families, societies, and nations worldwide face social, economic, and cultural challenges due to the aging population (United Nations Population Fund, 2012). In the Philippines, the population of older citizens is also growing. The need for health care services continues to grow as the population ages. Indeed, the health of older adults is getting worse, especially when it involves degenerative and chronic diseases that are brought about by the body's immune system aging (Faner & Chiong, 2020).

To address this issue, the researchers propose developing an Android-based medicine reminder application for elderly individuals. The app features medication notifications that alert them to their prescribed schedule, prompting them to take their medicine. Furthermore, the Application includes a dependent feature that notifies family members or caregivers of the medication times for elderly individuals.

Creating an accessible and easy-to-use reminder application aims to support elderly individuals and their families. The study was a step toward helping elderly individuals to live independently and safely, in addition to being

a reliable option. It can help not only elderly individuals and families but also the health sector to address the issue and focus on medication adherence, which can be a lead to increase the positive rates of health matter of elderly individuals.

Project Objective

This project aims to develop android-based medicine reminder for elderly individuals. Therefore, Voice Notification was integrated to improve medication adherence. Furthermore, the project specifically aims:

1. To develop an android-based medicine reminder application that assists the elderly in prescribed medication intake using voice notification.
2. To incorporate a navigation feature that provides the location of nearby healthcare facilities for elderly individuals.
3. Evaluate the developed project to substantiate the accessibility and quality of the mobile application based on the ISO 25010 software standard.

Project Purpose

Medication adherence is indeed highly prevalent among elderly people. Many old people are facing the issue of complex medication schedules that are associated with memory loss and vision impairments. Prescribed time represents a significant medication regimen, which may result in worsening health issues, increased hospitalization rates, and life-threatening conditions.

Due to the problem faced by elderly individuals with rising rates of medication non-adherence, existing medicine reminder applications have significant limitations. Some applications lack an elderly-friendly design due to having complicated interfaces and small fonts that are difficult for the elderly with visual or cognitive impairments. Additionally, they offer limited voice

notification integration, depending mostly on text-based alerts that may be difficult to understand for the elderly who are having trouble reading. Furthermore, current systems do not provide GIS-based health facility navigation, causing users to not have location-based assistance for locating nearby pharmacies and healthcare facilities. The other significant gap is a lack of caregiver monitoring features, with most systems failing to provide immediate notifications reminding caregivers when medications are missed.

To deal with the problem, the project seeks to provide a solution by developing an Android-based reminder application that delivers voice notifications for medication names and indicates the prescribed time, which is different from the traditional reminder methods. Moreover, a notification for the people who are dependent on the user was implemented to notify family members, caretakers, and healthcare providers if the patient does not take the medicine at the prescribed doses or does not adhere to the scheduled medication time, ensuring that the elderly people get the needed support to stick to their treatments. The Android-based reminder application for elderly individuals can be beneficial to the following:

The primary beneficiary of this project is for elderly individuals by providing a more accessible and efficient way to manage their prescribed medication. Voice notification delivers audible reminders for better assistance, reducing reliance on traditional reminder methods that may be difficult for elderly individuals.

Family members and caregivers also benefit from this application, as they are allowed to (remotely) monitor medication adherence. By receiving real-time alerts when an elderly individual misses a scheduled dose, family

members and caregivers can provide timely assistance or reminders without constant physical supervision. It can help them monitor medication without the need for physical appearance or being beside elderly individuals.

Healthcare providers gain advantages from the application as well, as improved medication assistance for adherence leads to better treatment outcomes, lower hospitalization rates, and more effective patient care. Ensuring that elderly patients follow their medication treatment on prescribed medication schedules, doctors and pharmacists can make more accurate assessments for more efficient treatment and make necessary adjustments to prescriptions and to the changing digital world.

Future researchers and developers benefit from this study by citing this research, which helps them know what they can help to improve in the current study and application. This research serves as their reference for more insightful ideas and understanding of the methods employed. The findings helps future researchers to gather more ideas and essential reference points, guiding researchers entering the same study or same field and supporting the development of improved and innovative solutions in related research areas

Scope and Limitations of the Study

This project focuses on developing a mobile application to assist with the medication schedule intake using a mobile-based application. The primary beneficiaries of this study are elderly individuals to help them manage their medication schedules effectively. The researchers also provide GPS navigation that allows the user to locate nearby pharmacies, clinics, and hospitals in case they are needed. The application includes the following features and modules, such as:

Information Module. This module displays the details once the user sets their medication. It offers information regarding the type of medication, frequency, and the date and time for which they need to receive voice notifications. It enables the user to view information in an organized and structured manner. Furthermore, this module features a notepad that allows the user to easily jot down notes relating to their personal health matters. In addition, it includes a calendar feature to remind them of specific dates for doctor appointments and other personal events.

GIS Mapping Module. This module combines GPS technology with mapping services specifically designed for elderly individuals, enabling them to locate nearby pharmacies and hospitals when necessary. Utilizing this module assists users who may not know their precise location, particularly when they are at home. It aids users in finding facilities that offer support for their medication needs or health-related issues.

Notification Module. This module delivers audible notifications using text-to-speech technology (TTS) to remind users for a designated duration. It allows for medication reminders to be communicated in a clear and accessible way, benefiting users, particularly those with visual impairments or those who favor spoken alerts. Additionally, this module features acknowledgment functions that alert family members or caregivers if the user consistently neglects their medication reminders.

Data Analytics Module. This module provides a system for monitoring and analyzing medication adherence. The purpose of this system is to give caregivers and users an overview of medication intake records, which track the history of doses taken or missed based on the date selected by the user. It also

have a graph so that the user can easily see not just the date but the percentage and count of doses missed and taken. The important component is the medicine stock feature, which can help users manage the inventory of medicines and alert users if stock needs to be refilled. To provide more overview information, it also has a reminder set count that shows how many reminders are set by the user.

Dependent Module. This module enables the monitoring of the user's medication information by authorized family members, caregivers, or healthcare providers. It helps the elderly individuals who may require additional support to manage their health independently. The system provides real-time alerts for dependents to be notified when the user misses the medication intake without the need for constant physical appearance. This module gives access to the user's medication list and schedules, promoting better health monitoring by understanding the user's medication habits.

Database Module. This module serves as the central data repository for the entire application. It is built to secure all stored data entered by the user. It includes data storage and retrieval to protect all information of the application features. Besides storage, it is in charge of ensuring that data integrity and security are applied to protect the information of users and prevent unauthorized access.

User Account Management Module. It serves as the main gateway to use the application, responsible for handling user registration, authentication, and profile management to ensure that data is protected and secured, allowing individuals to manage their authorized accounts.

Emergency Contact Module. This module shows the essential

emergency hotlines within their area and personal emergency contacts, such as family members or caregivers. It supports critical situations, such as medical emergencies, accidents, or other critical events, to help them assess and handle health-related matters and situations.

Limitation of the Study

The mobile application has several weaknesses that limit its overall usefulness in assisting elderly healthcare management. The primary shortcoming lies in the medication reminder feature, which lacks specific categorization of medicines, preventing a systematic arrangement of multiple prescriptions. Furthermore, the application does not provide real-time traffic updates, which makes it difficult for caregivers and elderly individuals to plan visits to healthcare facilities efficiently. The mapping feature is further limited because it fails to recognize newly established hospitals or smaller local pharmacies, preventing immediate access to critical healthcare services.

Moreover, the current location of elderly users cannot be tracked by caregivers through the map, reducing the application’s ability to ensure safety and provide timely assistance during emergencies. The system also lacks complete contact information for hospitals and clinics within the mapped area, which may delay urgent medical consultation. The medication reminder feature faces linguistic limitations, resulting in improper pronunciation of certain medicine names, which may confuse elderly users who rely on auditory notifications. Caregivers are also unable to monitor whether elderly users follow prescribed medication schedules, limiting the application’s effectiveness as a comprehensive management tool that can be helpful for families and caregiver to monitor the location of elderly individuals.

CHAPTER II

THEORETICAL FRAMEWORK OF THE PROJECT

Review Related Literature

This chapter provides a thorough review of related literature and studies sourced from reliable materials such as online articles, journals, books, and other credible publications. It serves as a foundational guide for the researchers in conducting the study. Additionally, it analyzes and connects various studies to enhance the understanding and significance of the research.

Elderly Individuals

Aging is a natural and inevitable growing phenomenon that affects and changes the psychological, physical, hormonal, and social conditions (Neuhaus & Neuhaus, 1982). According to (Birren & Renner, 1997), aging is a biological phenomenon as a composite of natural underlying mechanisms that occur in mature organisms as they grow old under normal environmental conditions. While aging is an unavoidable part of life, it is often seen as an arduous period in many ways, and it is. As individuals age, old age is considered a major challenging phase in life. They become more dependent on others for guidance and support. Due to old age, physical activities are reduced, financial stability may decline, and their life is more vulnerable due to their condition.

Elderly individuals face the need to interact, and having an interaction with others can help them to have relationships and connections with other individuals. Elderly individuals can join and support groups that are connected to associations. As part of these groups, most elderly individuals believe that it makes their routine different, especially in rebuilding social connections and decreasing loneliness (Leite et al., 2002). Responsibility is shared between the

family and the state for the care services of elderly individuals. Demographic change in family results in a ‘beanhole structure,’ which implies that in multiple generations of great-grandparents and grandparents, it is necessary for them to receive support and help, but a single offspring is also a challenge for them (Reichert & Phillips, 2008).

Elderly individuals need constant attention. they need love, understanding, and care from human beings, especially from their families. It is a duty to see that they are not suffering and spending the twilight years of their life in misery, penury, and isolation. Vital support is an important aspect of life to improve the overall quality of life of human beings, especially for elderly individuals. Old age homes have become essential in today's fast-paced world, where family dynamics are shifting and the traditional responsibilities towards elderly individuals are fading. As a result, crucial support is expected to be shown to sustain their lifestyle and quality of life (Kivelt and Scott, 1979).

Elderly individuals often forget to take their medicine on schedule, and it can be a challenge for them to remember which medicine should be taken at the prescribed time. There is a need for people to assist patients who are taking medications, reminding them consistently of the prescribed time of their medication. (Sree, et al., 2020).

Medication Adherence

Accuracy with medication involves many factors, including the patient, healthcare providers, system intervention, and individual treatment (Jimmy & Jose, 2011). Health outcomes and costs can be reduced by ensuring that prescribed medications are taken at the appropriate time and with the correct dosage (Schiff et al, 2003). This has been shown to improve medication

adherence. Adherence to medications leads to lower treatment costs, faster recovery times, a decrease in disease progression, and fewer hospital stays (Shapiro et al, 2017).

Poor adherence is particularly problematic in older individuals, as the likelihood of experiencing multiple conditions and more regimen complexity increases with age (Marangoni et al, 2011). However, elderly individuals are not immune to these factors. Care providers face a significant challenge in meeting the medication requirements of elderly patients. The elderly frequently rely on drugs to improve their quality of life, increase their lifespan, and prevent or treat illnesses. However, older adults often do not follow their prescriptions as directed, and this has adverse clinical and economic effects (Gellad et al, 2011).

Poorly controlled medication adherence leads to increased symptoms and disease severity, complications, and potential mortality that can harm society (Chisholm-Burns & Spivey, 2012). Older adults' adherence to medication is influenced by various factors such as the patient's experience, medications used, healthcare providers provided at the time of diagnosis, healthcare system functioning, and socioeconomic factors. However, patient medication adherence is essential in guaranteeing therapeutic benefits (Yap, 2016). Interventions that help the improvements in medication are being targeted at reminding patients to take their medication or refill prescriptions on time (Guthrie, 2001).

The dose is only effective if it is taken at the precise time with the correct quantity; if not taken correctly, it could potentially worsen the patient's health (Das, 2022). Interventions that help the improvements on medication

adherence are being targeted at reminding patients to take their medication or refill prescriptions on time (Guthrie, 2001). However, many interventions to improve medication in the elderly found that the use of reminders on medication packaging, reminder calendars, and mailed reminders did not improve medication adherence over control groups (Wolters et al., 2015).

Medication Management

Medication management is a procedure carried out by medical therapists, the goal of which is to maximize therapeutic outcomes for patients. Many medicines have specific dosages that need to be taken at regular intervals. Missing dosages or taking medications at the wrong time can lead to medical complications. Medication management includes interventions such as sending reminders to patients to take their medications, filling pill boxes, and labeling the lid on each compartment to designate when they need to be taken (Mohammad et al, 2019).

Medication is a key component of the treatment of many diseases, between 20% and 30% of prescriptions fail to be filled, and around half of those are not taken. An estimated $100 billion to $290 billion in preventable medical expenses are incurred each year due to these adherence gaps (Peterson et al., 2003). Patients often refuse to take their prescribed medicine for many reasons, such as poor practices in prescribing that create complex regimens, concerns about side effects and cost, mistrust in the benefits of medication, and lack of knowledge in health (Beseecher et al. 2007).

Elderly people's daily lives are made more complicated by the burden of administering medications, particularly when complex instruction regimens are involved. This added responsibility can additionally influence their ability to

manage medication (Maidment et al., 2020). Additionally, as people age, they lose some cognitive skills, such as processing speed as well as certain memory, language, visual, and executive functions (Sluggett et al., 2020). According to a study that evaluated problems with medication self-management amongst older adults with and without visual impairment, over 29% of patients over 65 needed help managing their medications, even with the use of visual aids (McCann et al., 2012).

Numerous environmental factors, including social factors like family and caregiver support and the home environment, should be considered when evaluating different barriers to medication-taking because research suggests that feedback from care partners and the home environment can affect older adults' ability to self-administer medication (Faisal et al., 2020). Medication adherence apps are becoming more interactive and customizable, it remains unclear how much these features enhance medication or if they help maintain user engagement over extended periods (Chen et al., 2016) One of the newest forms of healthcare treatments that can effectively support the health of older persons is mobile health, or mHealth. Features including event reminders, alarms, and messaging are included in mobile health apps. These properties make it one of the most often utilized technology-based solutions for managing drug use since they may be used to overcome challenges including forgetfulness (Pratiwi et al., 2023). Mobile apps are seen as a promising method to connect with underserved populations, as they have the potential to enhance access to healthcare, improve quality, and bolster social support particularly among vulnerable and hard-to-reach communities (Anderson-Lewis et al., 2018; Peek, 2017).

Medications can reduce distressing symptoms, prevent acute and chronic illnesses, and improve the health of patients. Medication therapy management is an essential service that provides goals and features. Medication management may lessen some medication-related problems like non-adherence and costs in the usage of health care, but the evidence is not enough to support the improvement in health outcomes (Viswanathan et al., 2015).

Voice Notifications

Voice notifications use speech synthesis technology to deliver messages in spoken form, making it easier for users to understand important information quickly. This system is especially useful for emphasizing critical notifications, ensuring they are noticed and acted upon. By converting text into speech, voice notifications reduce the need for users to read messages, saving time and improving the efficiency of information delivery (IntBell, n.d.).

However, many people are getting sick, especially the elderly. As they take their medicine, many elderly struggles with taking medications correctly. Often forgetting to take or taking the wrong amount (Mehala & Gripsy, 2020). People used an alarm just to remind them to take their medicine. And some studies provide solutions like automatic pill intake notifications that remind users when it is time to take medication through flashes, beeps, text messages, and phone calls (Jinto & Chakraborty, 2023). A study published in the Journal of the American Medical Association (JAMA) on December 2, 2024, revealed that text message reminders for patients who delay refilling their medications did not improve medication adherence over the course of a year. Using voice notifications, elderly people help cue to take scheduled medications, making

use of the voice notifications as a reminder system, as it may be more effective than text messaging (Okuboyejo & Olugbara, 2017). Based on the study published in JMIR Formative Research, the use of voice assistant reminders among older adults managing chronic pain was investigated. The study found that participants responded to voice reminders with an average latency of 55 minutes, indicating that while voice notifications prompted medication intake, the timing varied among individuals (Mahoney et al., 2021). Additionally, research featured in BMC Medical Informatics and Decision Making evaluated an IVR system designed to remind patients about medication refills. The pilot study demonstrated that participants found the voice-based reminders acceptable and feasible, suggesting potential benefits in enhancing medication adherence (Houston et al., 2008).

GIS-Based Mapping

The use of Geographic Information Systems (GIS) for analysis and assessment features the importance of research in services for elderly people. Roger Tomlinson introduced GIS technology in 1963, which indicates that it has grown significantly and been used widely in many different fields over the years. Progress in GIS research has been found effective over the last 50 years, demonstrating its potential as a flexible tool and its growing recognition as essential in modern applications for daily life (Jozi et al., 2010). Understanding the geographic distribution of medical resources and how it affects availability and accessibility is one important area where GIS has proven useful (Lu et al., 2019). The availability of healthcare resources has a significant impact on the well-being of older individuals, and the distribution of social services is directly impacted by the location of these resources (Talen & Anselin, 1998).

It is difficult to access the availability of essential facilities and services from the quality of life lived by the elderly. On the one hand, however, the elderly population is dispersed and has limited accessibility. Accessibility to a variety of services and facilities becomes an important concern for older adults with limited mobility (Liu & Engels, 2012). As a result, study and research on the accessibility of urban community services and amenities must be conducted (Wang et al., 2021). Due to a higher probability of having several severe chronic diseases and more frequent illnesses, older adults consume prescription drugs more than any other age group (Kantor et al., 2015).

Therefore, having easy access to a pharmacy is important for the overall health of elderly individuals. Community pharmacies offer a variety of services in addition to distributing prescription drugs, such as screening tests, immunizations, health promotions, patient counseling, and educational programs (Melton & Lai, 2017). According to previous studies, pharmacists can reduce healthcare costs while enhancing medication safety, promoting health, preventing disease, improving medication adherence, and enhancing patient outcomes when they consult with doctors and other medical professionals (Spears, 2010).

The large number of visits indicates that pharmacies in the community are easily accessible by medical experts who usually interact with patients (Berenbrok et al., 2022). In addition to being essential for keeping drug supplies available, pharmacies also play a crucial part in society by offering quick access to healthcare for everyone (Walton, 2004). Communication between pharmacists and patients improves medication adherence and treatment effectiveness, resulting in better health outcomes, according to a systematic

review (Rajiah et al., 2021). The integration of GIS in healthcare evolves the measurement of spatial access to healthcare services, resulting in more effective resource allocation calculations. GIS approaches have been used extensively to create the most effective solutions to allocation challenges as technological innovation evolves (McLafferty, 2003).

Synthesis and Relevance of the Study

The reviewed studies focus on elderly individuals, medication adherence, medication management, voice notifications, and GIS-based mapping in healthcare. These topics illustrate the significant challenges faced by elders in managing their healthcare services. With aging comes physical deterioration, memory loss, and increased reliance, making proper medication intake and access to pharmacies more challenging.

As technology continues to grow, mobile applications, voice reminders, and GIS mapping are being studied as viable ways to support elderly care and improve health outcomes. Research consistently shows that proper medication management and timely reminders can significantly improve adherence and lower health risks. Voice notifications appear to be more effective than traditional reminder methods like pill organizers and text messages, which have mixed outcomes. In addition, GIS-based studies emphasize the importance of spatial access to healthcare services, as accessibility directly affects medication availability and continuity of treatment. The findings show that combining voice reminder systems with location-based healthcare services can improve the overall health management of elderly individuals. The integrated technologies promote independence among elderly users by reducing their dependence on caregivers for daily health-related tasks. The use of GIS further

supports informed decision-making by identifying areas and improving healthcare planning.

Table 1. Matrix of Related Literature and Studies

Authors	Title	Year	Findings
Jimmy, B., & Jose, J.	Patient Medication Adherence: Measures in Daily Practice	2011	Medication adherence is one of the main factors that help to improve health outcomes and avoid complications. Being irregular in taking medications leads to more frequent hospitalizations and, in turn, a rise in healthcare costs, which, in turn, becomes a valid need for effective intervention strategies.
Marengoni, A., Angleman, S., & Melis, R.	Aging with Multimorbidity: A Systematic Review of the Literature	2011	Older people often cope with several chronic diseases at once, which in most cases require complex medication regimens. Ideas such as simplified reminders and tools for the proper management of their medicines could help these patients.

Schiff, G. D., Fung, S., Speroff, T., & McNutt, R. A.	Decompensated Heart Failure: Symptoms, Patterns of Onset, and Contributing Factors	2003	Keeping track of the medicines the patient takes in line with the schedule of the prescribed medications makes the patient more likely to avoid readmissions, and also, the treatments are more likely to be successful. Besides medicines, digital reminders are looked upon as a very practical tool in the support of the anticoagulants administered to the patient at home.
McCann, R. M., Jackson, A. J., Stevenson, M., Dempster, M., McElnay, J. C., & Cupples, M. E.	Help Needed in Medication Self-Management for People with Visual Impairment: Case-Control Study	2012	A considerable number of elderly patients, specifically the ones with visual losses, have trouble with the management of their drugs. This is why voice alerts have become quite popular, as they can reach out to people with disabilities.
Fenerty, S. D., West, C., Davis, S. A., Kaplan, S. G., & Feldman, S. R.	The Effect of Reminder Systems on Patients' Adherence to Treatment	2012	Systems using both sound and visual notifications have proven to be better adherence tools for

			patients. The elderly seem to like interactive and personalized approaches the most and thus tend to show 100% compliance.
Faisal, S., Ivo, J., McDougall, A., & Patel, T.	Stakeholder Feedback on Electronic Medication Adherence Products: Qualitative Analysis	2020	Research has found that voice through mobile applications and caregiver support has the most significant effect on medication adherence in older people. Personalized tips assist in user participation and regularity in medication administration.
Pratiwi, H., Kristina, S.A., Widayanti, A. W., Prabandari, Y. S., & Kusuma, I. Y.	A Systematic Review of Compensation and Technology-Mediated Strategies to Maintain Older Adults' Medication Adherence	2023	Consequently, mobile health applications also contain voice reminders, another treatment with great efficacy found in many places that helps older adults administer medicines on time, and thus, the number of missed doses due to forgetfulness is brought down.

Jayasudha, A. R., & Krishnan, A. M.	Medicine Reminder Through Voice Guide Using IoT	2024	Voice-guided medication reminders incorporated for completion adherence are largely triggered by personalized follow-up alerts that remind seniors of their medication at the right time. Through this study, researchers discovered technology as an assistive tool for old people whose health is already weakened.
Sree, B. V., Indrani, K. S., & Latha, M. S. G.	Smart Medicine Pill Box Reminder with Voice and Display for Emergency Patients	2020	Tests conducted by the “medicine adherence study” have demonstrated that the use of combined visual and voice-based alerts allows better adherence in the elderly, in particular those who have trouble with memory and mental issues.
Ali, A., Naveed, J., & Hamza, M.	Voice Reminder Assistant Based on Speech Recognition and Speaker Identification Using Kaldi	2021	Technologies like voice assistants are developed to be user-friendly. The visualization of the voice-based instructions notifications on the app helps people with cognitive

			impairments read it out using the remote control. Thus, it enables elderly individuals and people with disabilities to read it even if they cannot read it properly.
Anderson-Lewis, C., Darville, G., Mercado, R. E., Howell, S., & Maggio, S.	mHealth Technology Use and Implications in Historically Underserved and Minority Populations in the United States: Systematic Literature Review	2018	Health applications in the context of mobile technology (mHealth) deliver an expedited medicine supply to elderly individuals who suffer from restricted movement or have limited healthcare support.
Maidment, I., Lawson, S., Wong, G., Booth, A., Watson, A., & Bailey, S.	Towards an Understanding of the Burdens of Medication Management Affecting Older People: The MEMORABLE Realist Synthesis	2020	For elderly patients, it is increasingly demanding to manage their medications, mainly because of their cognitive decline, complex dosing schedules, and the lack of help from the caregiver.
Wolters, M.	Using Voice Assistants for Elderly Care	2015	Voice-activated reminder systems alternatively raise the level of adherence to medicine in people aged 65 and over, with most of them being tech-

			challenged and showing a dislike for traditional mobile or text-based notifications.
Viswanathan, M., Kahwati, L. C., Golin, C. E., Blalock, S. J., Coker-Schwimmer, E., Posey, R., & Lohr, K. N.	Medication Adherence Interventions: Comparative Effectiveness and Implementation	2015	The study reveals that personalized voice notifications and mHealth solutions are the two main factors that are a good guarantee that patients are taking their meds in the prescribed time.
Gellad, W. F., Grenard, J. L., & Marcum, Z. A.	A Systematic Review of Barriers to Medication Adherence in the Elderly: Looking Beyond Cost and Regimen Complexity	2011	Factors like forgetfulness, cognitive decline, and the unavailability of easy-to-use medication reminders of the kind accessible to the elderly exert a negative impact on the level of adherence to the drugs among older adults.

Guthrie, R. M.	The Effects of Reminder Systems on Medication Adherence in the Elderly	2001	Automatic reminding systems, especially those using voice alerts, reported a lack of missed doses in the elderly by approximately 40%, as those people would be reminded by voice.

Conceptual Framework

Figure 1 shows the conceptual framework of the application and how it works by connecting its modules. It follows the Input-Process-Output (IPO) framework with integrated feedback mechanisms to support elderly users and their dependents in maintaining medication adherence.

The input phase begins when users access the application through the user account management module, where they register, log in, and manage their profiles. After working on their account, users can set their medication schedules. take notes about their health-related thoughts, schedule appointments, and add emergency contact information for caregivers, family members, and healthcare providers.

The process phase involves systematic data handling and manipulation through interconnected modules that work together to transform input data into meaningful actions. Once medication information is entered, the system processes it to create personalized medication schedules based on prescribed frequencies and timing. The notification module processes medication schedules and triggers reminders in both audible and visual formats. The

dependent module processes notification updates to ensure that its dependents are informed in real time when medications are taken or missed.

The output of the system includes timely visual and voice medication reminders, updated notifications sent to dependents, medication intake adherence reports, current medicine stock status, accessible emergency contact information, and maps that locate nearby healthcare resources such as pharmacies, hospitals, and clinics; provide navigation assistance; and enable quick access during emergencies or routine pharmacy visits.

Feedback loops occur when the elder and their dependent respond to the outputs by confirming medication intake; the data analytics module analyzes and feeds insights back into the notification module to adjust reminder timing based on user response patterns and trigger alerts to dependents when doses are missed. These updates are saved back into the database as new inputs. This framework shows how each module works together to improve medication adherence, safety, and support for elderly individuals.

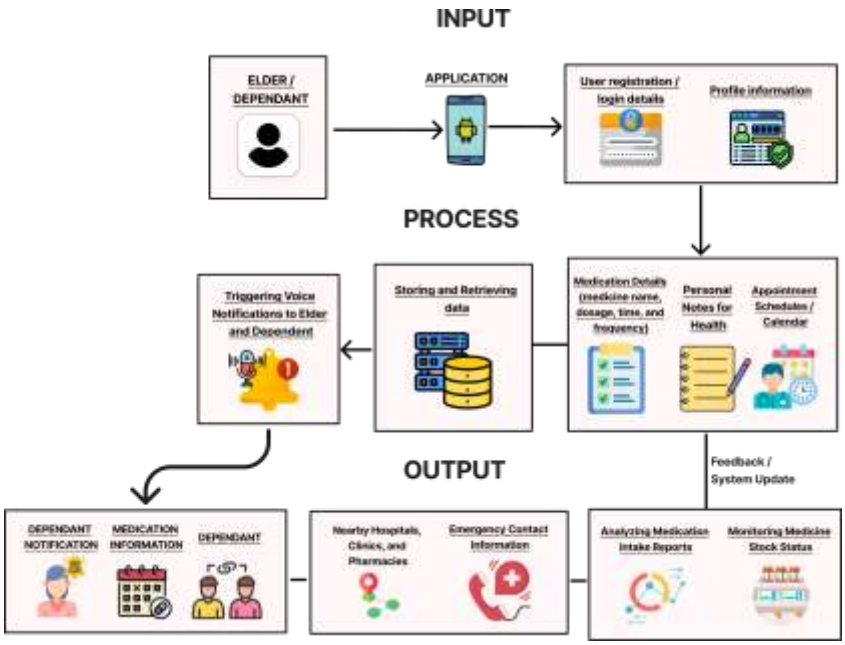


Figure 1. Conceptual Framework

CHAPTER III

METHODOLOGY

This chapter discusses all the methods and processes used in this study to develop the proposed mobile Application designed to assist elderly individuals with their medication adherence. The researchers used various approaches and strategies to guide the project's development, all of which contribute to creating a system aimed at improving medication compliance among elderly individuals.

Project Design

Figure 2 illustrates the architecture of an Android-based medicine reminder system. The user interface, which is the main access point to all functionality, is where the system begins. To access their information and create a personal account, users can register and log in. The medication management features allow users to add, edit, and delete medicines from their list as needed. Users are reminded via alerts from the reminder notification feature, helping them take their prescribed medicines as intended and ensuring consistent medication adherence. The system includes a navigation feature that allows users to find nearby hospitals, pharmacies, and clinics. The notes acts as a recording feature that stores users' important health information.

Additionally, users can view and monitor their medical appointments through the appointment scheduling feature. In addition, if a user skips or misses a dose, a dependent module notifies family members or caregivers. A centralized database stores all data generated by these features, ensuring it is accurate and secure across the system. Lastly, the user can view their

medication inventory and alert the dependent if elderly individuals are in an emergency.

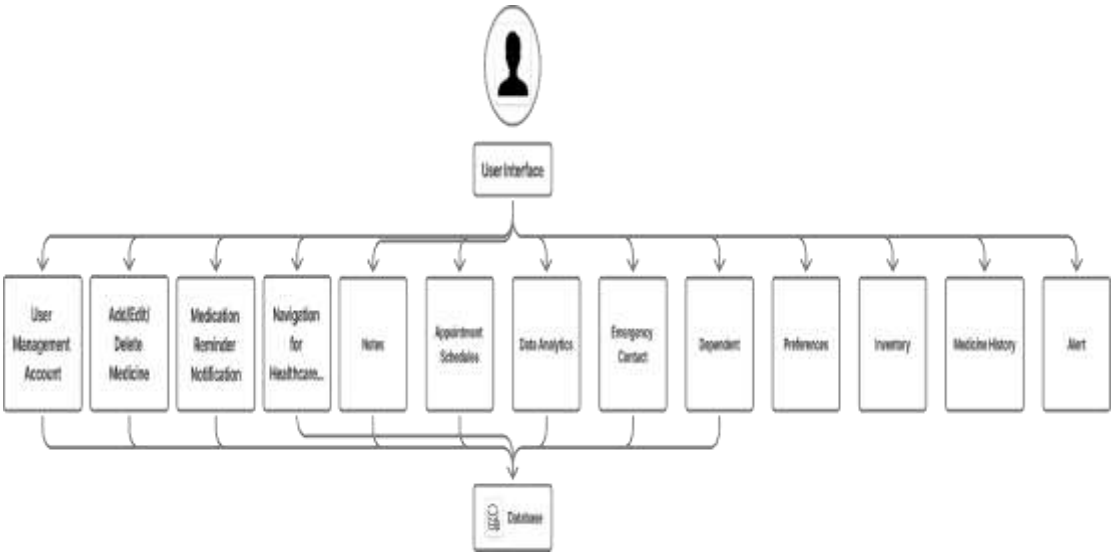


Figure 2. System Architecture

Project Locale

The study was conducted by researchers in Los Baños, Laguna, a Philippine municipality known for its rural and urban areas, nearby healthcare facilities, and growing elderly population. The location was chosen because it is relevant to the project's target audience, which consists of elderly individuals who often have difficulty taking their medications as prescribed due to memory impairment, a lack of supervision, and limited access to easily understandable digital resources.

Moreover, the selected locale allowed the researchers to observe real-world conditions affecting elderly medication adherence, such as limited caregiver availability and varying levels of digital literacy. These factors provided valuable insights in evaluating the usability and practicality of the developed application within the target community.

Project Development

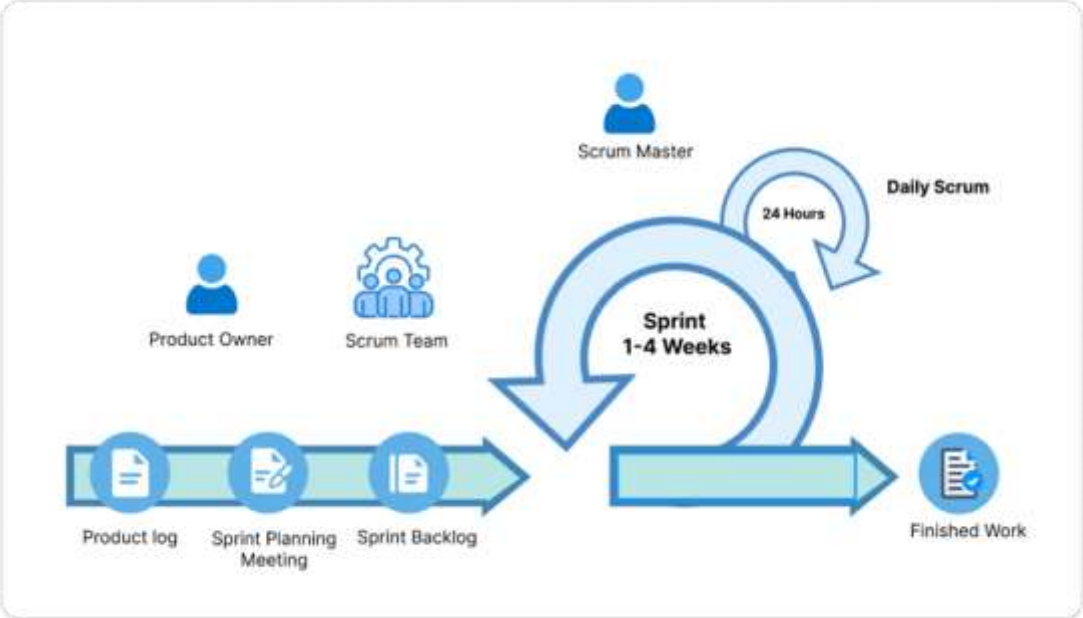


Figure 3. Agile Scrum Methodology

The development of the Android-based medicine reminder application followed the Agile Scrum methodology to ensure user-focused, flexible, and iterative progress. Roles are distributed among researchers to assist in developing an Android-based medication adherence application.

It is the responsibility of the product owner to organize and select essential features, such as an easy-to-use interface for users, emergency alerts, medication prescription schedules, and, especially, voice notification, which is highlighted as the most significant part of medicine reminder application development. By analyzing the scrum model's activities, the scrum master addresses problems such as technical challenges, collaboration, and uncertain requirements to improve the process. The Scrum team designs and develops location-based mapping, medication reminders, and dependent reminders. It is improved and enhanced by developers, designers, and testers.

Every sprint begins with a sprint planning meeting, where the team

selects the most essential features from the product backlog, then operates and develops them for 1 to 4 weeks to design and test them. A daily scrum helped track progress, address blockers, and align efforts. The Sprint Backlog included actionable items such as voice notification development and usability testing. At the end of each sprint, a functional and testable version of the system was produced, ready for feedback and evaluation by elderly users and their caregivers to ensure the app's efficiency and usability.

Product Backlog

The product backlog is a prioritized list of what needs to be accomplished to enhance the product. It helps the team identify priority tasks and allocate resources effectively, encompassing all the necessary functions and enhancements required to fulfill the project. The team regularly updates the product backlog in response to user feedback, team performance, and project objectives. This ensures that all development activities remain aligned with the scopes and timeline.

In addition, the product backlog serves as a central reference for monitoring development progress and ensuring transparency among team members. Each backlog item is clearly defined to guide implementation and testing activities during each sprint. This approach helps minimize misunderstandings and reduces the risk of overlooking critical system requirements. The prioritization of tasks also allows the team to focus first on features that deliver the greatest value to end users. Through regular backlog refinement, the researchers ensure that the system evolves in response to both technical evaluations and user needs ensuring consistency with project objectives and timelines.

Table 2. Product Backlog (Elderly Individual)

PRIORITY	USER STORY	ESTIMATED TIME
1	As a user, I want to create an account	1hr
2	As a user, I want to login to the system.	1hr
3	As a user, I want to add and set my medication intake at the exact time and day.	3hrs
4	As a user, I want to see the list of the medicines that I added.	2hrs
5	As a user, I want to edit or delete the information of the added medicine if needed.	2hrs
6	As a user, I want to write down all my health-related information.	1hrs
7	As a user, I want to set a reminder notification for my schedule of checkups and other health-related appointments.	2hrs
8	As a user, I want to know what is the nearest pharmacy, Hospitals, and clinics in my area.	3hrs
9	As a user, I want to edit my personal information.	3hrs
10	As a user, I want to view other features in settings	3hrs

The table above shows the product backlog for the elderly. Each item is organized by the importance of its functions and features, such as setting the medication schedule, adding notes, and setting appointment reminder schedules.

Table 3. Product Backlog (Dependent)

PRIORITY	USER STORY	ESTIMATED TIME
1	As a dependent, I want to create an account	1hr
2	As a user, I want to login to the system.	1hr
3	As a dependent, I want to see the list of medicines that the user takes.	2hrs
4	As a dependent, I want a notification when the user is done taking their medicine.	1hr
5	As a dependent, I want to view the medicine inventory of the user.	1hr
6	As a dependent, I want to set a reminder notification for my schedule of checkups and other health-related appointments.	2hrs
7	As a dependent, I want to know what is the nearest pharmacy, hospitals, and clinics in my area.	3hrs
8	As a dependent, I want to edit my personal information.	1hr
9	As a dependent, I want to view other features in settings.	2hrs

The table shown above is the product backlog for the dependent. It prioritized functions and features such as viewing the user's medicine inventory and the list of medicines. The projected time for each product is included in the product backlog.

Sprint Backlog

The sprint backlog organizes and oversees development tasks during the sprint, ensuring the most critical items are prioritized. Tasks are organized by value and urgency, along with the estimated effort required to complete them.

Table 4. Sprint Backlog (Elderly Individual)

PRIORITY	USER STORY	ESTIMATED TIME
1	As a user, I want to create an account.	1hr
2	As a user, I want to login to the system.	1hr
3	As a user, I want to add and set my medication intake at the exact time and day.	2hrs
4	As a user, I want to see the list of the medicines that I added.	2hrs
5	As a user, I want to edit or delete the information of the added medicine if needed.	2hrs
6	As a user, I want to write down all my health-related information.	2hrs

The table above lists the sprint backlog for the elderly. The scrum team collaborates in sprint planning. Any changes are communicated in the team's

scrum. These tasks guarantee that the prioritized functions and features are organized on delivering the relevant features for the end-users.

Table 5. Sprint Backlog (Dependent)

PRIORITY	USER STORY	ESTIMATED TIME
5	As a dependent, I want to view the medicine inventory of the user.	1hr
6	As a dependent, I want to set a reminder notification for my schedule of checkups and other health-related appointments.	2hrs

The table above is the sprint backlog of the dependent. The priorities that the scrum team needs to accomplish are shown above. It includes the product's functions and features. It ensure the scrum team focuses on the task at hand.

Sprint Planning

The researchers hold weekly meetings to assess the project's progress and determine whether any modifications or new features are necessary. The team analyzes the system's performance during these sessions to identify challenges and obstacles encountered throughout the sprint and to explore potential solutions. The sprint planning meeting's goals are to improve project outcomes and ensure that projects are proceeding properly.

System Flowchart

The flowchart illustrates the process of the Android-based medication reminder app. The procedure starts by accessing the Application, where users can either sign up for a new account or log in with their existing credentials. Users may choose their account type, either as an elderly individual or a

dependent, which also enables users to manage their profiles. Primary users are the elderly who receive medication reminder notifications. The caregiver or family member who serves as a dependent should be notified if their chosen elderly user marks the scheduled dose as taken or skipped.

Users are directed to the dashboard, where this function can manage a medication list, find nearby healthcare facilities using the map, create and view health-related notes, set appointment notes, and view analytics of the users' data that show their medication use and medicine stock. To end their interaction with the system, users can log out of their accounts in the application settings. The functions listed in the figure can help support medication adherence and minimize the risk of missed doses.

Additionally, the application allows caregivers to monitor adherence patterns, providing an overview of medications taken or missed over time. The map features not only shows nearby healthcare facilities but also updates information about hospitals, clinics, and pharmacies to assist in urgent situations. Health-related notes and appointment reminders are synchronized across accounts, ensuring both older people and their caregivers remain informed. Voice notifications are customizable, allowing reminders to be set according to the user's preferred language and volume. Analytics dashboards visually represent medication intake trends, making it easier for caregivers to identify potential issues. The application also provides alerts when medicine stock is running low, reducing the risk of missed doses due to unavailability or out-of-stock conditions. Overall, the system's integrated features aim to improve medication adherence, promote timely medical visits, and enhance

communication between elderly users and their caregivers. These overall features are beneficial for the health system to improve using technology.

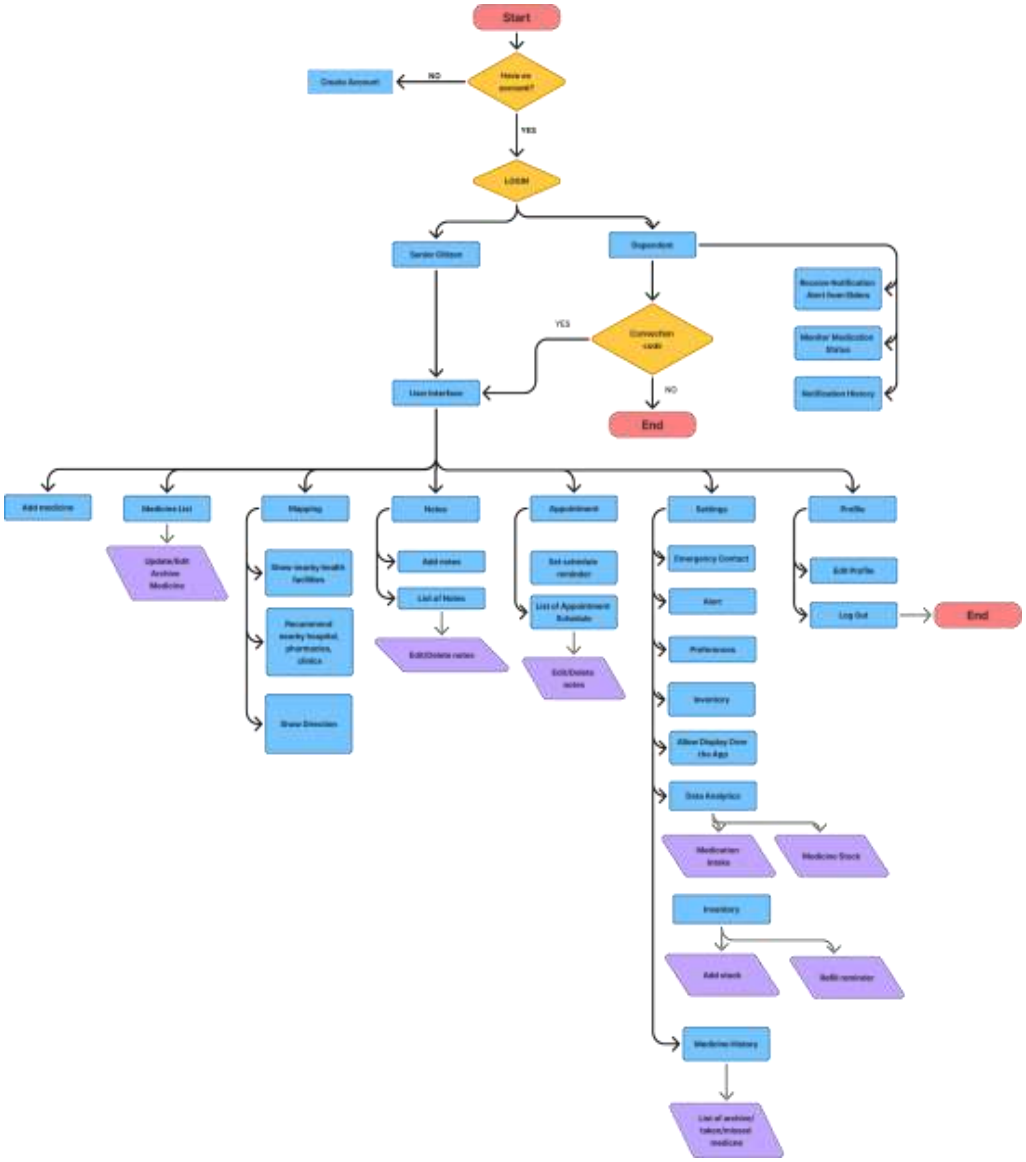


Figure 4. Flowchart

Use Case Diagram

Figure 5 illustrates the use case diagram for the proposed Android-based medicine reminder application. The system supports two main user roles: elderly individuals and dependents.

The elderly individual user is the primary beneficiary of the system. They can register or log in, add medications, view their medication schedule, and receive timely medication reminders. In addition to enhancing medication management, the system includes features to set appointment schedules, add personal health notes, view nearby healthcare facilities such as pharmacies, clinics, and even hospitals on the map, mark medications as taken, and update profile or account settings.

The dependent is the caregiver or family member who can also register or log in, manage medications, view schedules, set appointments, and access notes. If the elder forgets to mark the medication as taken, the dependent were notified to take their prescribed medicines. This feature ensures that the dependent stays informed and can provide timely assistance. It ensures that both users participate in managing and monitoring medication, enhancing safety and communication between the elder and their dependent. The use case diagram supports the project's purpose and objective: fostering connections and support among caregivers and family members of elderly individuals, especially to improve medication adherence.

The user account management team is responsible for managing user access for both elderly individuals and dependents, allowing users to create and manage their authorized accounts to ensure their data is stored securely. For dependents, this feature works if elderly individuals or users accept the supervision of their selected dependents. Once permission is granted, dependents can view and manage the elderly individual's medication information and health data without accessing the elderly individual's account.

A dependent feature makes the medication more useful for the elderly, as it helps them track their family’s or an elderly individual's medication regimen.

The data analytics tool provides valuable insights into medication adherence for both the elderly and their dependents. The system collects data about medication regimens and whether medications have been marked as taken. This data is analyzed and turned into visual reports and summaries that are accessible to both users. These reports provide the elderly with an overview of their adherence records, which can be beneficial during meetings with medical professionals.

For the dependent, the analytics dashboard offers an in-depth review of elderly medication adherence statistics over time. This feature helps the dependent to identify potential problems, such as an ongoing sequence of missed medications. The emergency contact is designed to provide immediate assistance in critical situations by providing contact information for hotlines and the trusted person, typically a dependent. The system includes reminders for coming medical visits, lessening the possibility of missed check-ups and encouraging regular health monitoring. Notifications can be customized for frequency and timing to meet the elderly user's daily schedule. The software also supports synchronization across several devices, allowing dependents and caregivers to access updated information immediately. The map feature is regularly updated to reflect new healthcare facilities and pharmacies, improving accessibility and emergency planning. Overall, these extended features reinforce the app’s goal of promoting medication adherence, improving communication between elderly users and dependents, and providing reliable support for elderly healthcare management.

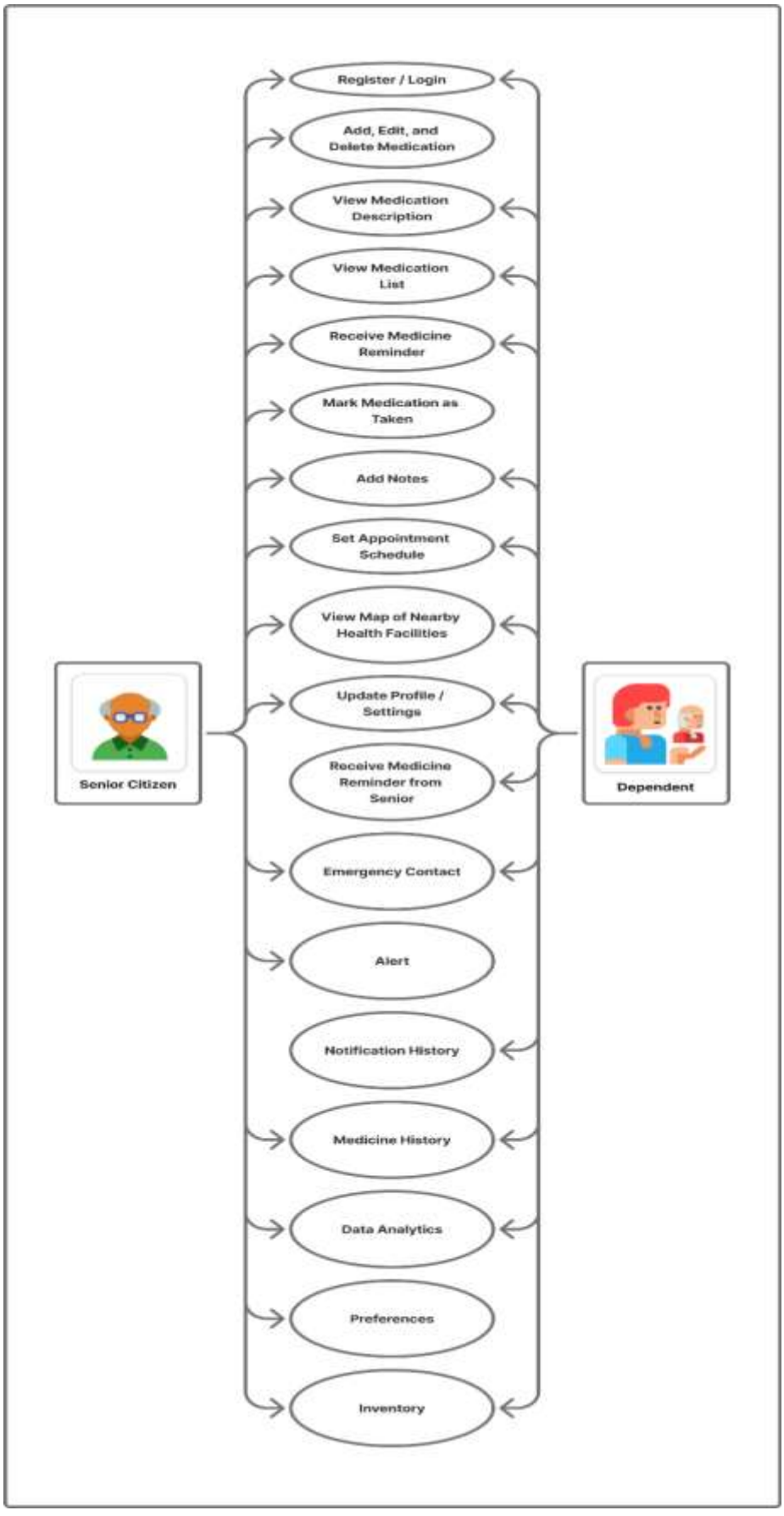


Figure 5. Use Case Diagram

Geographical Information System

MapBox was used and integrated to support the project's Geographical Information System. This integration allowed the Application to display the map. It improves the system's capabilities to provide directions for navigating a location. The GIS component of the Application utilizes multiple data layers to provide a comprehensive navigation experience. These include a base map layer, a user location layer, health facility markers, and a route layer for navigation. By combining these layers, the Application enables elderly individuals to easily locate and navigate to the nearest hospitals, pharmacies, or barangay health centers, improving their access to essential medical services. Providing real-time mapping and location tracking optimizes the Application's usability and assists users who may require access to prescription medication or urgent medical assistance.

Materials

The researchers utilized a set of materials and tools to create and finalize the mobile Application. Materials are critical to the system's construction, as they significantly influence its functionality, performance, and usability. Choosing suitable materials and technologies is essential to guarantee that the Application operates efficiently, is user-friendly, and remains cost-effective. By thoughtfully selecting tools, platforms, and development resources, the researchers successfully executed the project's objectives, ensuring that the final product meets the intended design specifications experience for its users.

Software

Software is essential for creating the functionalities and features of this Application. It consists of the instructions or programs that allow a computer to

execute specific tasks. The software development process includes designing, testing, and coding to verify the operation of the program, if it is correct and fulfills user requirements.

React Native

React Native is an open-source framework for building mobile apps, especially for Android, that lets you create real, exciting mobile apps with JavaScript alone. It includes libraries, reference materials, and pre-built components. It helps developers to develop user-friendly mobile applications that display user interface features.

Typescript

TypeScript (TS) is a high-level programming language that adds static typing with optional type annotations to JavaScript. It is designed for developing large applications and transpiles to JavaScript.

Mapbox API

The Mapbox API allows developers to integrate mapping functionality into their applications. It provides access to a wide range of mapping tools and features, including custom map styles, geocoding services, and route optimization.

OpenStreetMap

OpenStreetMap (OSM) is a collaborative project that creates a free, editable map of the world. It is maintained by a community of volunteers who collect data through surveys, aerial imagery, and other free sources. The platform is widely used for various applications, including navigation, and data visualization, allowing developers and researchers to freely access, and integrate geographic information into web and mobile applications.

Visual Studio Code

Visual Studio Code is an open-source editor that facilitates efficient and streamlined coding for developers across different platforms. This platform offers tools for code refactoring, navigation, and deep semantic code comprehension.

Firebase Database

Firebase is a powerful platform that enables developers to create, manage, and expand applications effortlessly while offering a variety of tools and services to support the development, management, and growth of their mobile applications. Firebase provides backend services that enhance the accuracy and security of app development, including the Realtime Database, Cloud Firestore, authentication, and hosting.

Gradle

Gradle is an open-source automation tool designed to simplify the development process of software projects. It includes building, testing, and deploying applications across various platforms and programming languages

Hardware

The hardware used to develop, test, and implement the Application is listed below. Each component was chosen to ensure system compatibility, simulate real-world user conditions during testing, and ensure seamless performance upon final deployment. By testing the application with these hardware components, developers can ensure the system runs continuously, produces accurate notifications, and remains responsive across a variety of real-world situations and capable of properly managing medical services for elderly individuals.

Hardware	Specification
RAM	8GB
CPU	HP/ Intel Core i5-8365U CPU@ 160GHz
OS	WINDOWS 10
STORAGE	256GB

The table above outlines the specification requirements for a PC or laptop to operate the developed system efficiently and without issues. These specifications are essential for assessing the overall performance and potential of a computer system. These are the recommended specifications for the laptop/PC as suggested by the researchers.

Hardware	Specification
RAM	4GB
ANDROID OS	Android 11
PROCESSOR	OCTA CORE
STORAGE	64GB

Table 7. Hardware Specification for Mobile Phone

The table outlines the particular specification requirements for mobile phones intended for testing the application to support the development of the system. These specifications ensured that the application could be evaluated under conditions and compatibility across devices.

Project Testing and Evaluation Software Testing

The software designs provided by the researchers are shown in the figure above and adhered to during the testing process. These phases involved verifying and testing the developed architecture and assessing its user-friendliness and the thoroughness of its functionality.

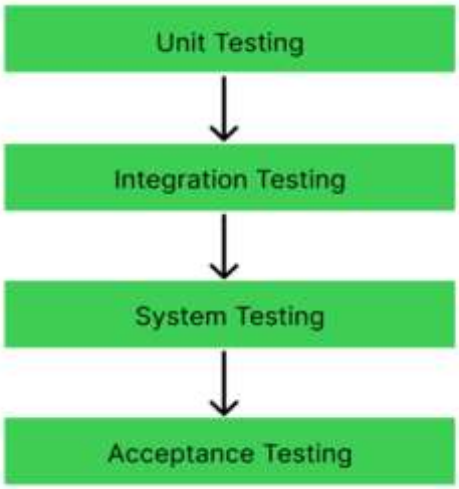


Figure 6. Software Testing

Unit Testing

Unit testing is a software development approach that focuses on evaluating code to identify problems and issues within the system. The purpose of this testing is to allow quality assurance to investigate the behavior of the system. This method helps prevent minor bugs from evolving into major structural failures during later stages of the project.

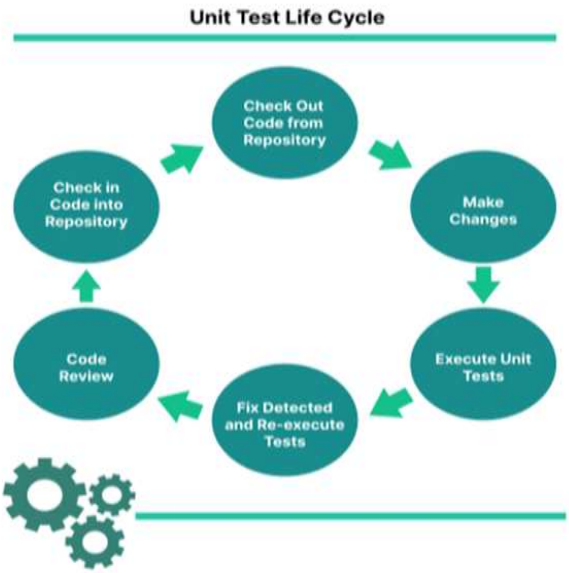


Figure 7. Unit Testing

System Testing

The figure shows a conceptual model of the system testing process; it demonstrates how inputs, events, and system operations lead to the expected outcomes based on the specified requirements. The main aim of this review is to pinpoint the crucial functions that are necessary. This assessment of the system focuses on its operational outcome, performance compatibility, and reliability.

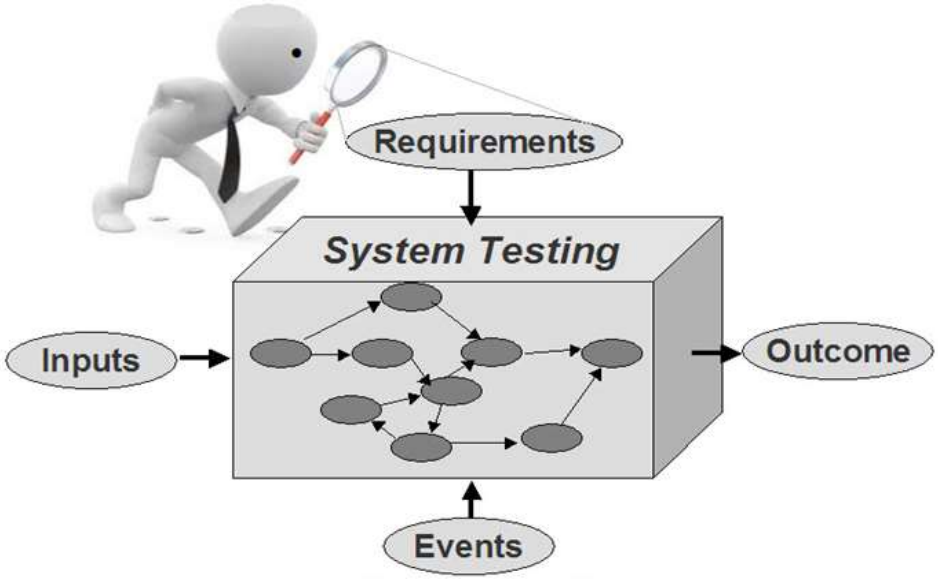


Figure 8. System Unit

Integration Testing

The integration testing consisted of a systematic assessment process that examined the functionality and features according to the design. This is important to guarantee that every function operates correctly and interacts appropriately with one another. Through this process, any inconsistencies, errors, or conflicts between modules are identified and resolved, ensuring that the application performs reliably, efficiently, and as intended before deployment.

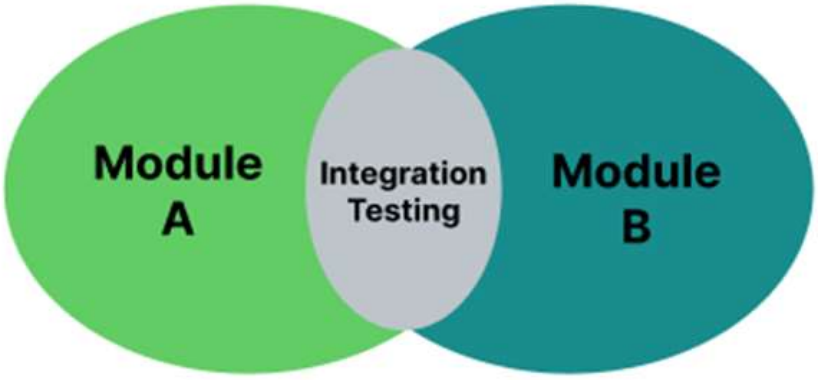


Figure 9. Integration Testing

Acceptance Testing

The figure shows the process of acceptance testing. It involves examining the completed project to verify that it fulfills the specified requirements, this undergoes a series of evaluations to assess its design and how well it meets user expectations. The researchers carry out formal assessments based on the user's requirements to ascertain if the system is satisfactory for the user. During this process, each function and feature is tested against the initial project specifications to ensure completeness and accuracy of the application. Any discrepancies or areas requiring improvement are documented and addressed before final approval, guaranteeing that the system meets both technical standards and user satisfaction.

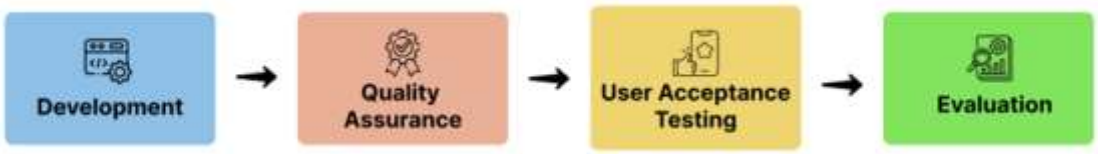


Figure 10. Acceptance Testing

Evaluation Procedure

Figure 10 illustrates the visual presentation of the ISO 25010 product quality model. The ISO 25010 standard is a widely recognized standard for software product quality to evaluate the system. This evaluation focuses on the system's functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability. ISO 25010 provides systematic approaches in evaluating system design. This assessment is conducted by research developers to help to identify potential problems or issues for improving the system design and its implementation.

Functional suitability is an important aspect, as it pertains to how it fulfills the user's needs and provides the necessary functions. Performance efficiency is crucial for determining the efficiency of the system to determine if it utilizes resources and meets performance requirements and its capacity. Compatibility relates to the system's ability to simply exchange information with other systems and use data from external sources. Usability ensures that the system is user-friendly and understandable to the user. Reliability assesses the quality of the system and its accuracy by analyzing and determining the system's quality. Security focuses on protecting sensitive information and data and managing access to authorized personnel. Maintainability assesses the system if it requires maintenance and adaptability to its environment. Portability addresses the system capability of its function across different platforms. It also provides a set of qualities against which stated quality requirements can be compared for completeness to implement solutions to maintain consistent performance across platforms.

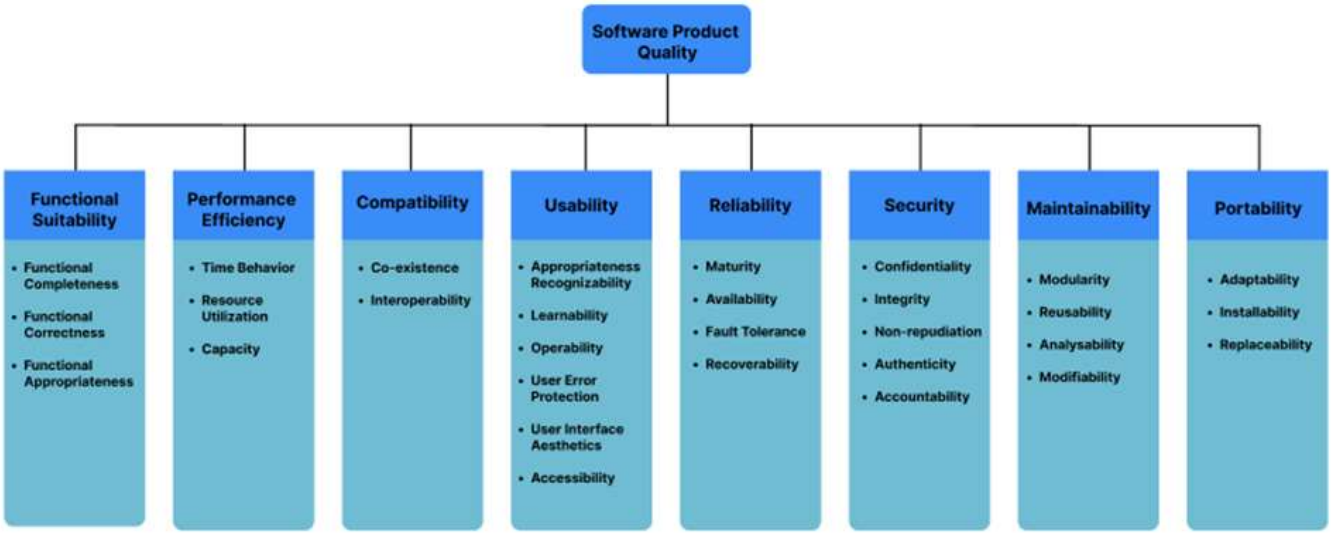


Figure 11. Software Product Quality

Population of the Study

The population of the study consists of the 50 respondents for Technology Acceptance Model and 3 IT professionals who evaluate ISO 25010 and test the system. These individuals are the target respondents for the questionnaire, as they have direct interaction with the system. Their feedback is crucial in evaluating the system's overall performance for the areas that need improvement.

The table below shows the IT professional respondents who analyze and evaluate the reliability of Android-based systems, including the performance of developed features. It provides insight and recommendations to enhance and improve the Android-based system.

CATEGORY OF RESPONDENTS	SAMPLE
User	50
IT Professional	3
Total	53

Sampling Design

The design is a purposive method that analyzes individuals from the selected area to participate in the evaluation based on the research assessment. A variety of individuals are selected, including elderly individuals, caregivers, and family members, with a total of 50 respondents in the community. This approach ensures that the category of individuals who uses the application is included in the study. Additionally, participants in the IT professionals have a total of three (3) individuals. This approach focuses on the assessment of the system’s performance and functionality with the expertise relevant to the system.

Data Collection Instrument

A structured questionnaire was used as the primary instrument of researchers to gather data, which is implemented to evaluate the Android-based medication reminder application's overall performance, usability, and efficiency. The goal of the questionnaire was to collect responses from the system's target audience, which consists of elderly individuals, their family members or caregivers, and IT professionals who tested and examined the application.

The researchers prepared hard copy surveys to be used for respondents who don't have internet access, ensuring that everyone is included in the data collection procedure. Additionally, the research survey instrument includes Likert scale questions to determine participant's perceptions, experiences, and insights regarding the system to support the research's stated goal of evaluating system acceptance and identifying areas for improvement.

Statistical Instrument

Researchers use population sampling as a purposive sampling method to gather and analyze the data from respondents. To ensure the accuracy and transparency of the data from respondents, the researchers use the Likert scale as a survey technique to assess the user feedback. There are four potential responses to a series of questions: four (4)—Strongly Agree, three (3)—Agree, two (2)—Disagree, and one (1)—Strongly Disagree. This approach enables an in-depth evaluation to ensure the reliability and accuracy of the data collected.

Table 8 shows the Likert scale used in the survey to assess respondents' data. The evaluation of respondents is measured by Mean and SD. This statistical analysis allows the researchers to quantify user perceptions and determine the overall satisfaction level with the system. The combination of purposive sampling and the Likert scale ensures that the feedback collected is both relevant and representative of the target population, supporting data-driven conclusions for the study. It provides a clear understanding of how well the application meets user expectations and highlights areas that may require improvement.

Table 8. Likert Scale Technique

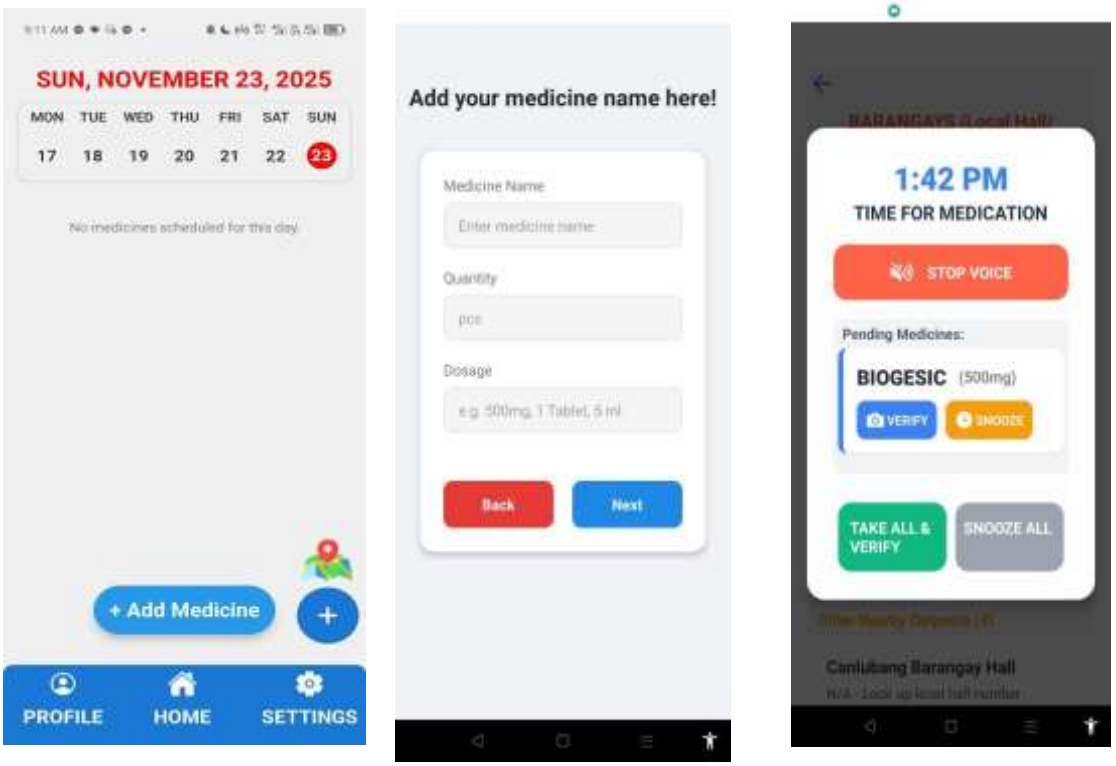
RATING	MEAN RANGE	INTERPRETATION
4	3.25 – 4.00	Strongly Agree
3	2.50 – 3.24	Agree
2	1.75 – 2.49	Disagree
1	1.0 – 1.74	Strongly Disagree

CHAPTER IV

RESULTS AND DISCUSSION

This chapter presents the testing, evaluation, and interpretation of the results of the developed system in relation to the objectives of the study. The focus is on the conducted and planned assessments of the system's functionality, integration, and quality standards.

Android-based medicine reminder application that assists the elderly in prescribed medication intake using voice notification.



(a) Medicine Reminder List (b) Medicine Form (c) Reminder Notification

Figure 12. Android-based medicine reminder application

Figure 12 displays the Android-based medicine reminder application, including the reminder list, form, and notification. It assists users in adding their medicine name, dose, schedule, and reminder frequency. The notification

appears at the scheduled time, showing the details of the medicine the user has set to view their medication on the application screen.

A navigation feature that provides the location of nearby healthcare facilities for elderly individuals.

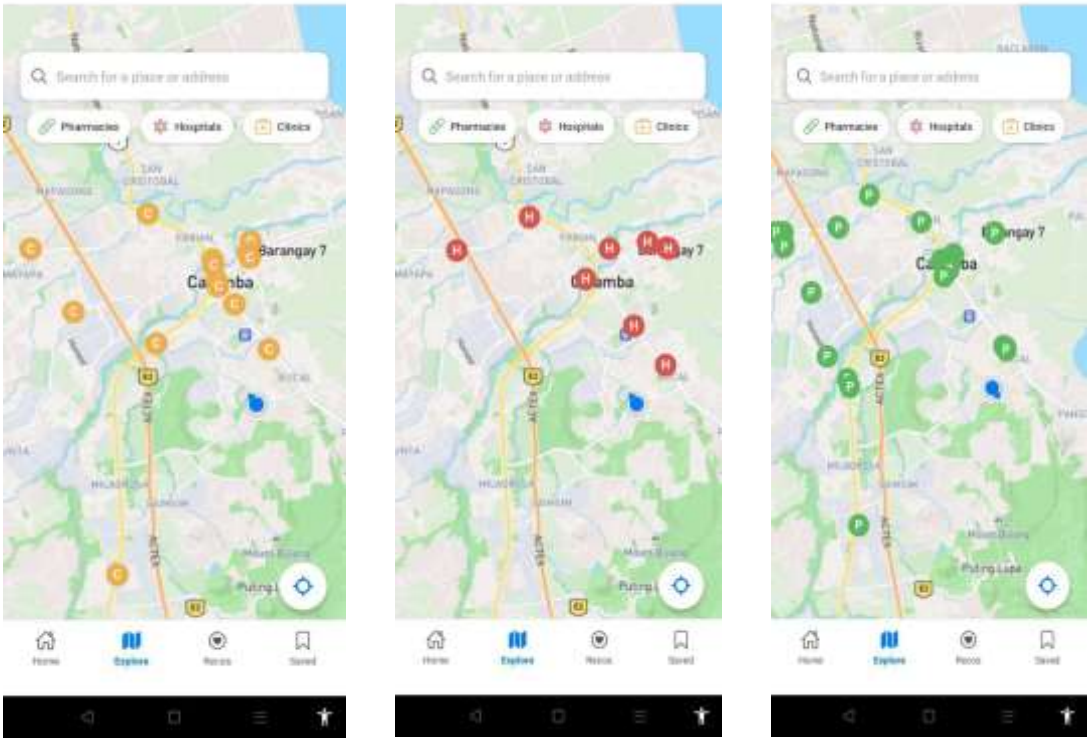


Figure 13. GIS-Mapping feature

Figure 13 Maps includes a mapping feature to help users locate nearby healthcare facilities such as hospitals, clinics, and pharmacies. Users can see their location and search for a specific destination. React Native was utilized as the development platform, and MapBox is used to make the mapping feature work as well as for the tiles and the location's direction.

UNIT TESTING

Table 9 presents the results of the unit testing for the Information Module. It indicates that the actual output is functioning correctly in accordance

with the expected output. This ensures that the data handling and its smooth interaction between the user and the system are seamless.

Table 9. Unit Testing for the Information Module

TESTED CASE	EXPECTED OUTPUT	ACTUAL OUTPUT	PROCESS	REMARKS
Add new medicine schedule with complete details	Medicine is successfully added and displayed on the dashboard	Medicine successfully added and displayed on the dashboard	2 seconds	Passed
Add medicine schedule with missing required fields	System displays error message: “Please fill in all required fields.”	System displayed error message: “Please fill in all required fields.”	2 seconds	Passed
Edit existing medicine schedule	Updated schedule is saved and reflected on the dashboard.	Updated schedule successfully saved and reflected on the dashboard.	3 seconds	Passed

The evaluation for the GIS Mapping module testing focused on the accuracy and responsiveness of features such as locating nearby pharmacies, hospitals, and clinics, as well as providing real-time directions from the current user’s location to selected facilities. By comparing the actual output and expected output, it ensures that the system performs correctly. The testing process also assessed the reliability of map updates, including the identification of newly established healthcare facilities and changes in location data. Usability was evaluated to ensure that both elderly users and dependents could easily navigate the mapping functions, contributing to the overall reliability and usability of the application.

Table 10. Unit Testing for the GIS Mapping Module

TESTED CASE	EXPECTED OUTPUT	ACTUAL OUTPUT	PROCESS	REMARKS
Load map on dashboard	Map loads successfully and displays the current user location.	Map loaded successfully and displayed the current user location.	3 seconds	Passed
Locate nearby hospitals, pharmacies, and clinics	System displays nearby pharmacies, hospitals, and clinics on the map with markers.	System successfully displayed nearby pharmacies, hospitals, and clinics with correct markers.	5 seconds	Passed
Get directions to selected facility	System provides real-time directions from user's location to selected facility.	Real-time directions successfully displayed on the map.	3 seconds	Passed

Table 11 presents the unit testing findings for the Notification Module. This serves as the heart of assisting elderly users in remembering their medications. The testing checks the functionality, speed, and accuracy of reminder alerts, such as displaying a medicine notification schedule by voice notifications and handling errors when there is no schedule set. Beyond just checking if notifications appear, the researchers evaluated whether the reminders are easy to understand for seniors who may struggle with complex digital interfaces. Comparing expected and actual outputs ensures that the notification system works properly and helps the user to take medicine at the scheduled time. The testing assessed the consistency of notifications across different devices and screen sizes to guarantee accessibility for all users. The responsiveness of alerts under varying network conditions was also examined to ensure reliability and guarantee accessibility for all users.

Table 11. Unit Testing for the Notification Module

TESTED CASE	EXPECTED OUTPUT	ACTUAL OUTPUT	PROCESS	REMARKS
Trigger scheduled notification	Notification is delivered on time with correct medicine details	Notification was delivered on time with correct medicine details	3 seconds	Passed
Multiple reminders	Notifications are delivered for all scheduled medicines without duplication	All scheduled notifications delivered correctly without duplication	5 seconds	Passed
No schedule set	System displays error message “No scheduled medicine”	System displayed “No scheduled medicine” message	2 seconds	Passed

The Data Analytics Module testing shown above evaluates how well the system processes and makes sense of the information that matters most to elderly users and their caregivers. It also evaluates the accuracy of processing and summarizing data, such as medication schedules, and monitoring whether the user skips their medication. It ensures that the system correctly analyzes user inputs, including statistical summaries. By comparing the expected and actual outputs, the results confirm that the module functions properly. During testing, the system was observed to generate correct summaries for daily, weekly, and monthly medication intake. Alerts related to missed doses are recorded accurately in the analytics dashboard. The module successfully handled multiple user accounts without errors, maintaining the integrity of each elderly user’s data. Overall, the module performed consistently under standard usage scenarios, confirming its reliability within the application framework.

Table 12. Unit Testing for the Data Analytics Module

TESTED CASE	EXPECTED OUTPUT	ACTUAL OUTPUT	PROCESS	REMARKS
Calculate missed medicines	System displays total number of missed medicine schedules.	System correctly displayed total missed schedules.	3 seconds	Passed
Display visual charts	System shows graphical representation (bar/line chart) of adherence trends.	Graphical charts successfully displayed with correct data.	3 seconds	Passed
Handle no data available	System displays message “No data available to analyze”.	System displayed “No data available to analyze” message.	2 seconds	Passed

Unit testing was conducted for the dependent module shown above. It emphasizes tracking the elderly's medications and ensuring that the system accurately keeps and corresponds to their accounts. Evaluating the predicted and actual results shows that the module is working as expected. The testing further confirmed that dependents could successfully access multiple elderly accounts once permission was granted. Notifications for missed or delayed medications are received promptly by dependent users. Visual indicators, such as adherence percentages and missed doses, are displayed accurately for each elderly user. The system correctly restricted access when permissions are not granted, ensuring the security of personal data. The tests demonstrate that the dependent module reliably supports caregivers in monitoring and managing the medication schedules of the elderly, reinforcing its functionality within the application. The seamless synchronization between the elderly user’s input and caregiver’s dashboard ensures real-time updates, accurate information sharing, and efficient coordination of care.

Table 13. Unit Testing for the Dependent Module

TESTED CASE	EXPECTED OUTPUT	ACTUAL OUTPUT	PROCESS	REMARKS
Assign medicine schedule to dependent	Schedule is linked and displayed under the dependent profile	Medicine schedule successfully linked and displayed	2 seconds	Passed
Edit dependent information	Updated details are saved and reflected in the dependent list	Updated details successfully saved and reflected	2 seconds	Passed
Alert Notifications	Receive notifications at the specific schedule base on elderly individual's medication intake time	Received a notification successfully based on elderly individual's schedule of its medication	3 seconds	Passed

The testing for the Database Module above verifies the accuracy of storing, retrieving, and updating records within the system. This module ensures that all data related to elderly individuals, dependents, schedules, and notifications are correctly managed in the database. Results show that the module handles data transactions reliably. Testing also confirmed that sensitive health information is stored securely, maintaining strict user confidentiality and data privacy standards. The system successfully executed rollback procedure that effectively preventing any potential data corruption. Overall, outcomes confirm that database provide robust and secure data storage, integrity, and reliability throughout system operations.

Table 14. Unit Testing for the Database Module

TESTED CASE	EXPECTED OUTPUT	ACTUAL OUTPUT	PROCESS	REMARKS
Add new elder record	Record is saved in the database successfully	Record is saved in the database successfully	2 seconds	Passed
Retrieve elder information	Correct information is displayed from the database	Correct information is displayed from the database	3 seconds	Passed
Invalid query request	System returns error or no record found	System returns error or no record found	2 seconds	Passed

User Account Management Module testing ensures that only authorized users can register, log in, and access the system. It ensured that the stored information was secure, encrypted passwords, and protected the user's information. The testing process carefully evaluated how the system handles user credentials during registration and login. Different user roles are tested to ensure that access permission was properly enforced. Attempts to access the system using invalid or unauthorized accounts are successfully restricted. The system password encryption mechanism was verified to prevent exposure of sensitive data. The module was also tested against multiple login attempts to prevent brute force attacks. Data validation checks are performed to ensure only correct information is accepted. The system maintained data integrity throughout all test scenarios. Audits logs are reviewed to confirm the user activities are properly recorded. Results indicate that the system effectively confirmed the authenticity of users who use the system. These tests helped ensure compliance with basic security standards. The results indicate that the

system effectively confirmed the authenticity of users who use the system. Overall, the module testing confirmed the reliability and security of the user account management features.

Table 15. Unit Testing for the User Account Management Module

TESTED CASE	EXPECTED OUTPUT	ACTUAL OUTPUT	PROCESS	REMARKS
User Registration	System accepts valid user details and creates a new account	Account successfully created with encrypted password	Input valid data (name, email, password) Submit Check database	Passed
User Login (valid)	System grants access with correct username and password.	User successfully logged in	Enter valid credentials Click login	Passed

Table 16 shows the result of the conducted unit testing for the Emergency Contact Module. This module provides quick access to hotlines at hospitals, pharmacies, and clinics, and it has stored contact numbers for the dependent, ensuring that help can be reached immediately in case of emergencies. These tests help ensure uninterrupted access to emergency services. The module was tested under different scenarios to confirm that it functions reliably even when multiple emergency contacts are accessed consecutively. The successful test results indicate that the module can support the safety and well-being of elderly users by providing timely access to vital services. The testing results show that it ensures that the module delivers accurate, accessible, and responsive emergency information. Table 16. Unit Testing for the Emergency Contact Module

TESTED CASE	EXPECTED OUTPUT	ACTUAL OUTPUT	PROCESS	REMARKS
Add emergency hotline (hospital, pharmacy, clinic)	Hotline number successfully saved and displayed in the contact list	Hotline number successfully saved and displayed	4 seconds	Passed
Add dependent contact (guardian/relative)	Dependent contact is successfully saved and shown in the list	Dependent contact successfully saved and displayed	3 seconds	Passed
Invalid contact input (e.g., letters in phone number)	System displays error message: “Invalid phone number.”	Error message displayed correctly	3 seconds	Passed

INTEGRATION TESTING

Integration testing was conducted to ensure that each of the developed modules would work properly together. The testing process focused on verifying the data flows correctly between interconnected modules. The interaction between the database and other modules was tested to ensure data consistency and integrity. It evaluates the interaction and communication among modules, such as GIS Mapping, Notification, Information, Database, Emergency Contact, Data Analytics, Dependent, and User Account Management Modules, to ensure the entire system provides a smooth and reliable user experience. Overall, the integration testing confirmed that the system operates as a unified and dependable platform.

Table 17. Integration Testing

TESTED CASE	EXPECTED OUTPUT	ACTUAL OUTPUT	PROCESS	REMARKS
Notification Module	The system allows user to receive a notification using voice, send notification to the dependent when elderly individuals skip its medicine.	The system successfully allows the user to receive a notification using voice notification, send notification to dependent when skip medication.	3 seconds	Passed
GIS Mapping Module	The system allows user to view the map, locate nearby pharmacies, hospitals, and clinics, and provide real time navigation.	The system successfully allows the user to view the map, locate nearby hospitals, clinics, pharmacies, provide real time navigation.	3 seconds	Passed
Information Module	The system displays the date, time, frequency and type of medicine details of scheduled medication, jot down notes related to its personal matters , and sets dates for its appointment.	The system successfully display date, time, frequency, and type of medicine details, jot down notes, and sets appointment.	3 seconds	Passed
Database Module	The system can stored data entered by the user, retrieves information.	The system successfully stored and retrieves data from database	3 seconds	Passed
Emergency Contact Module	The system can show the hotline for the emergency contacts.	The system successfully show the hotline for emergency hotlines	3 seconds	Passed
Dependant Module	The system can monitor of elderly individual's medication information.	The system successfully monitored elderly individual's medication information.	3 seconds	Passed
User Account Management Module	The system allows user to login, create account, and manage their data.	The system successfully allow the user to login, create and manage their account.	3 seconds	Passed
Data Analytics Module	The system can monitor and analyze the medication adherence, track and count history of doses taken or missed, and alert if the stock of medication need to be refilled.	The system successfully monitor, analyze, track, count history of doses taken or missed, alert if stock are need to be refilled.	3 seconds	Passed

SYSTEM TESTING

The system testing was conducted to evaluate the overall performance and functionality of the system. The test results show that all system modules performed as expected. The integration of the modules works well, and no problems encountered during testing. The system manages data management, sends notifications at a scheduled time, and locates nearby facilities on a map. It ensures that elderly individuals and dependents can interact with the system effectively. These results confirm that the system is working efficiently and delivering a positive user experience.

Table 18. System Testing

Tested Case	Expected Output	Actual Output	Process	Remarks
Users register and login	User redirected to dashboard after registration/login	Works as expected	2 seconds	Passed
Users add medicine schedule	Medicine saved and appear to dashboard	Works correctly	3 seconds	Passed
Notification triggered at scheduled time	Voice and text delivered on time	Delivered accurately	3 seconds	Passed
Dependent view elder's medication information	Dependent can view link elder's schedule	Displayed properly	4 seconds	Passed
GIS Map locate nearby pharmacies, hospitals, clinics and provide real time navigation	Map shows nearby facilities with direction	Accurate and responsive	3 seconds	Passed
Emergency contact holds information of hotlines.	Dial emergency hotline successfully	Works as expected	3 seconds	Passed
Database stores and receives user data	Data saved, retrieved, and displayed correctly	Stable	3 seconds	Passed

ACCEPTANCE TESTING

Test and Evaluate the Software using ISO 25010

Table 19. Test and Evaluate the Software using ISO 25010

Category	Mean	SD	Interpretation
Functional Suitability	3.78	0.39	Strongly Agree
Performance Efficiency	3.89	0.19	Strongly Agree
Compatibility	3.50	0.58	Strongly Agree
Usability	3.72	0.48	Strongly Agree
Reliability	3.50	0.29	Strongly Agree
Security	3.73	0.35	Strongly Agree
Maintainability	3.87	0.23	Strongly Agree
Portability	3.89	0.19	Strongly Agree
Composite	**3.74**	**0.34**	**Strongly Agree**

Legend: 1.00 - 1.74 = Strongly Disagree, 1.75 - 2.49 = Disagree, 2.50 - 3.24 = Agree, 3.25 - 4.00 = Strongly Agree

Table 19 presents the overall results of the ISO testing conducted on the developed system, indicating that the system meets ISO standards at an excellent level in most categories, as evidenced by a composite score of 3.74, which is interpreted as "strongly agree." The high scores for Performance Efficiency (3.89), Portability (3.89), Maintainability (3.87), Functional Suitability (3.78), Usability (3.72), and Security (3.73) demonstrate that the system performs its intended functions effectively, is accessible to elderly individuals, and properly secures information provided by users. Although all categories

received a rating of "Strongly Agree," the results also highlight areas that can be further enhanced. Compatibility (3.50) and Reliability (3.50) have the lowest means among the categories, indicating that there is potential for improvement in system stability across multiple devices and consistency as time passes. By evaluating the accessibility and quality of the mobile application using ISO 25010 standards, the identified areas for improvement suggest opportunities for future system improvement, which improve the application's capabilities and reliability over time.

Table 20. Evaluation of Perceived Usefulness by Respondents

Item	Mean	SD	Interpretation
The application helps me remember the time to take my medicine.	3.48	0.65	Strongly Agree
The voice notification is useful for medication reminders.	3.02	0.98	Agree
The map helps me quickly find the nearest hospital or pharmacy.	3.16	0.93	Agree
The Health Notes helps me record important information about my health.	2.96	0.88	Agree
Using the application contributes to better management of my health.	3.22	0.80	Agree
Composite	**3.17**	**0.85**	**Agree**

Legend: 1.00 - 1.74 = Strongly Disagree, 1.75 - 2.49 = Disagree, 2.50 - 3.24 = Agree, 3.25 - 4.00 = Strongly Agree

Table 20 demonstrates how respondents perceived the usefulness of the developed system. The statement on the application's ability to remind users of their prescribed medicines received the highest mean score (3.48,

Strongly Agree). The category had a composite mean of 3.17 (Agree), which implies that users mostly see the app as beneficial in helping them manage their health concerns. Meanwhile, the statement on the effectiveness of the Health Notes feature received the lowest mean (2.96, Agree), indicating that although helpful, this feature offers room for improvement.

Table 21. Evaluation of Perceived Ease of Use by Respondents

Item	Mean	SD	Interpretation
The application is easy to use and understand.	3.16	0.69	Agree
The voice notification is clear and easy to understand.	2.98	1.04	Agree
I can easily set the schedule for taking medicine.	3.43	0.61	Strongly Agree
The design of the application is orderly and simple.	3.46	0.61	Strongly Agree
I can easily find and use the map feature.	3.20	0.89	Agree
Composite	**3.24**	**0.77**	**Agree**

Legend: 1.00 - 1.74 = Strongly Disagree, 1.75 - 2.49 = Disagree, 2.50 - 3.24 = Agree, 3.25 - 4.00 = Strongly Agree

Table 21 shows how respondents assessed the application's ease of use. The simplicity and layout of the application's design received one of the highest ratings (3.46, Strongly Agree), indicating that the interface is acceptable to elderly users who prefer simple and uncomplicated displays. The estimated composite mean of 3.24 (Agree) indicates that users find the system simple and manageable for them to use. The clarity of the voice notification received the

lowest score (2.98, Agree), implying that some users may have encountered issues with volume, accent, or clarity.

Table 22. Evaluation of Attitude by Respondents

Item	Mean	SD	Interpretation
I enjoy using the app.	3.16	0.71	Agree
I feel confident while using the application.	3.10	0.93	Agree
Using the application helps me become more responsible.	3.32	0.78	Strongly Agree
The application makes managing my health easier.	3.33	0.77	Strongly Agree
Having all features in one app is a great idea.	3.42	0.77	Strongly Agree
Composite	**3.27**	**0.79**	**Strongly Agree**

Legend: 1.00 - 1.74 = Strongly Disagree, 1.75 - 2.49 = Disagree, 2.50 - 3.24 = Agree, 3.25 - 4.00 = Strongly Agree

Table 22 shows how respondents evaluated the attitude category, with a composite mean of 3.27 classified as "Strongly Agree", meaning that users value the application not only for its functionality but also for how effectively it works into their individual everyday lives. Many respondents appreciated having many health-related features combined into a single platform, reflected by the high evaluation of 3.42 for the particular claim. The item with the lowest mean, “I feel confident while using the application” (3.10, Agree), implies that some users may require additional practice or familiarity before using the app more independently. This findings suggests the importance of providing cleared

guidance to enhance user confidence and overall ease of use particularly among first-time or elderly users.

Table 23. Evaluation of Behavior/Behavioral Intentions by Respondents

Item	Mean	SD	Interpretation
I intend to continue using the application daily.	2.88	0.87	Agree
I would recommend this application to others.	3.60	0.61	Strongly Agree
I rely on this app for medication reminders.	2.94	0.84	Agree
I plan to use the app's various features daily.	2.94	0.90	Agree
I rely on this system to help maintain my health.	3.06	0.93	Agree
Composite	**3.08**	**0.83**	**Agree**

Legend: 1.00 - 1.74 = Strongly Disagree, 1.75 - 2.49 = Disagree, 2.50 - 3.24 = Agree, 3.25 - 4.00 = Strongly Agree

Table 23 shows that the behavioral intention results reflect how likely users are to continue incorporating the application into their daily lives. The highest score (3.60, Strongly Agree), showing that users recognize real-world value in the system and are comfortable recommending it. This classification obtained a composite mean of 3.08 (Agree), proving that the goal to use the system remains likely effective. Meanwhile, the intention to use the app daily has the lowest score (2.88, Agree), suggesting that usage frequency may vary depending on individual medication schedules or personal habits. This variation indicates that while users perceive the system as beneficial, consistent daily use may not be necessary for all users, particularly those with less frequent

medication requirements, this highlights the importance of flexible system features that can accommodate health management needs.

Table 24. Overall Result of the TAM Survey Answered by Respondents

Item	Mean	SD	Interpretation
A. Perceived Usefulness	3.17	0.85	Agree
B. Perceived Ease of Use	3.24	0.77	Agree
C. Attitude	3.27	0.79	Strongly Agree
D. Behavioral Intention	3.08	0.83	Agree
OVERALL WEIGHTED MEAN	**3.19**	**0.81**	**Agree**

Legend: 1.00 - 1.74 = Strongly Disagree, 1.75 - 2.49 = Disagree, 2.50 - 3.24 = Agree, 3.25 - 4.00 = Strongly Agree

The overall weighted mean of 3.19 (Agree) indicates that the system performs well in all measured areas. Respondents provided consistent views, shown by an overall SD of 0.81, suggesting the respondents experienced similar perceptions. The results are interpreted as Agree, illustrating that the application is beneficial, valuable, and appreciated, with a significant potential to succeed in daily usage. It implies that the system's effectiveness was perceived consistently across different user profiles. Positive outcomes reflect the success of voice notification in overcoming common accessibility barriers like visual impairment and cognitive decline. The data suggests that the system effectively addresses the specific need by simplifying complex medication regimens into manageable audible alerts, which adds a critical layer of security for family members and enhances confidence in the consistent and timely intake of prescribed medications.

CHAPTER V

SUMMARY, CONCLUSION, AND RECOMMENDATION

SUMMARY

The developed application enables users to create medication reminders to assist elderly individuals in managing their medications on time through a mobile platform. It includes critical functions such as health notes, appointment scheduling, data analytics, emergency contacts, and a mapping feature. These functionalities allow users to record health-related concerns, schedule check-ups and medication regimens, visualize medication adherence data, store important contacts for emergencies, and navigate nearby healthcare facilities such as pharmacies, hospitals, and clinics. These features give the elderly better access to and visualize their proper medication management.

The Agile Scrum methodology was used in the development of the system, covering the design process, development tools, and evaluation framework. The system was developed using TypeScript, React Native, JavaScript, Expo Go, Visual Studio Code, Firebase, and the Mapbox API. System evaluation was conducted using the Technology Acceptance Model (TAM) and ISO 25010 software quality standards.

In summary, the developed Android-based system successfully addressed medication management challenges and enhanced support for elderly individuals and their dependents. The system provides a more efficient, reliable, and user-accepted application that promotes proper medication adherence and improves overall healthcare assistance. The evaluation results showed a high level of user acceptance, indicating high usefulness, ease of use, and intention to use. Based on the ISO 25010 quality assessment, the

system achieved a high quality level in terms of functionality, usability, reliability, performance efficiency, and security. Moreover, the system earned a high-performance rating, indicating that the application performs efficiently in delivering timely notifications, processing data accurately, and maintaining stable operation during use.

CONCLUSION

The MED-ALALAY system was successfully developed to address medication non-adherence among elderly individuals in Los Baños, Laguna. The research accomplished its three primary objectives through careful development, implementation, and evaluation, creating an assistive healthcare support tool designed specifically for the needs of elderly people.

Develop an Android-based medicine reminder application that assists the elderly in prescribed medication intake using voice notification.

The researchers successfully developed Android-based medication reminder software featuring voice notification technology to help elderly individuals manage their prescribed medications. The application uses text-to-speech technology to provide audible reminders that read and speak the prescribed medicine and its schedule. To ensure immediate medication compliance, real-time notifications was sent to the family members and caregivers when elderly individuals miss their prescribed doses. This comprehensive approach not only improves medication adherence but also provide essentials peace of mind for families concerned about their love ones' well-being. Android-based medication represent a significant step forward in utilizing mobile health to enhance the overall quality of life for the aging population.

Incorporate a navigation feature that provides the location of nearby healthcare facilities for elderly individuals.

The application successfully incorporates Geographic Information System (GIS) technology via MapBox, providing older people and their families with tools to locating nearby medical facilities, checking place details, and receive informative step-by-step navigation directions. that helps elderly individuals, particularly those who may require immediate prescription refills or medical assistance.

Evaluation of the system application using the ISO 25010 software standard

The ISO 25010 evaluation showed that the MED-ALALAY system achieved a high software quality level, with a composite mean of 3.74 interpreted as "strongly agree" across all quality characteristics, including functionality, performance efficiency, usability, reliability, security, maintainability, and portability. User acceptance based on the Technology Acceptance Model (TAM) also indicated a high level of acceptance, with an overall weighted mean of 3.19 interpreted as “agree” and a “strongly agree” rating in users’ attitude toward the system (3.27). These results demonstrate that the MED-ALALAY system is a reliable, acceptable, and effective tool for promoting medication adherence among elderly individuals through its voice notifications, navigation, and monitoring capabilities.

RECOMMENDATION

Based on the findings of the study and identified limitations, future researchers are encouraged to consider the proposed recommended points for better enhancement:

Technical Enhancement

1. The system suggests that it needs to enhance the accuracy of locating places within the map. To ensure that all users in rural or urban-rural environments have access to essential healthcare services, creating a database of validated barangay health facilities, pharmacies, and clinics in the project area can be advantageous for the application to use 3D maps to enhance the user experience and locate broad local facilities.

UI/UX Enhancements

2. It is recommended to enhance the voice notification feature by allowing users to customize the tone and add language to make reminders more understandable, accessible, and engaging for elderly users who have hearing difficulties or language preferences. Using more comprehensive tools, applications, and software is beneficial for elderly users to have greater clarity in the names of medicines when voice notification is enabled.

Research Enhancements

3. Future developers should conduct usability testing with a larger population of elderly participants to obtain more accurate data. A regular assessment using ISO 25010 software quality standards is also recommended to ensure that the system consistently fulfills the proper standards of reliability, accessibility, and performance efficiency of the application.

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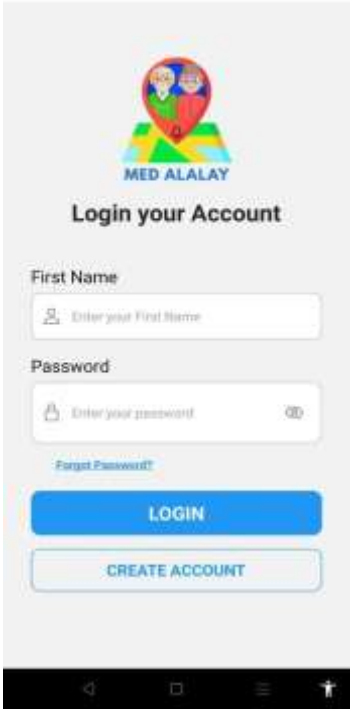
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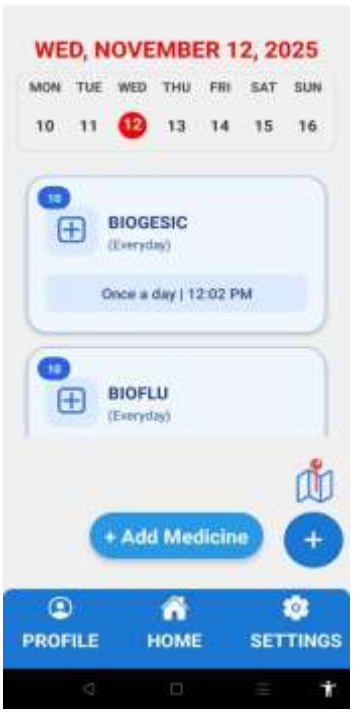
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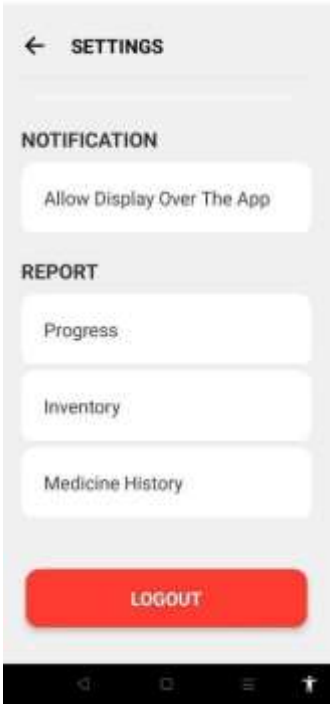
APPENDIX A.

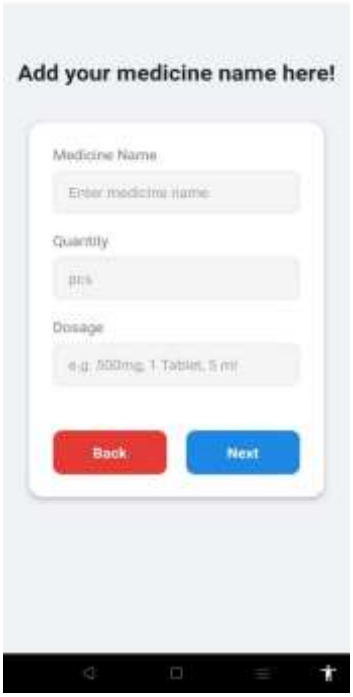
Sample Input/Output





The login page allows the user to access the application by signing in to their account. After the user clicks the sign-in, they'd be directed to the medicine reminder page, where they can add details about the medicines they need to take.

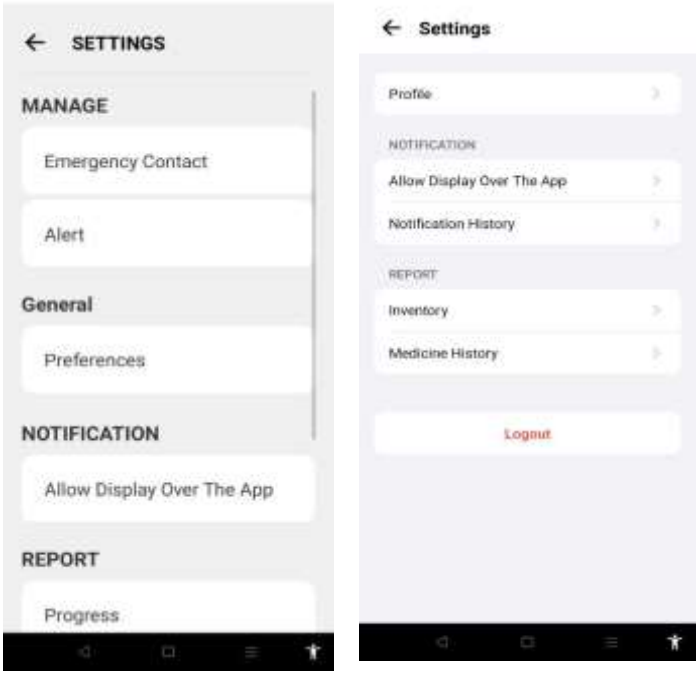




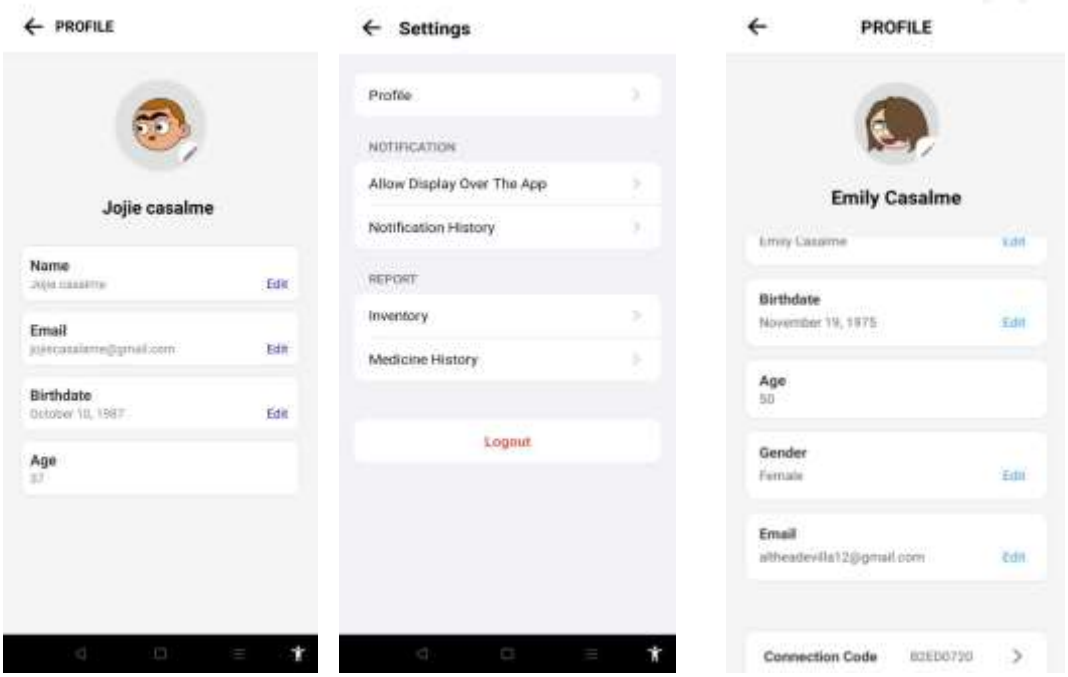




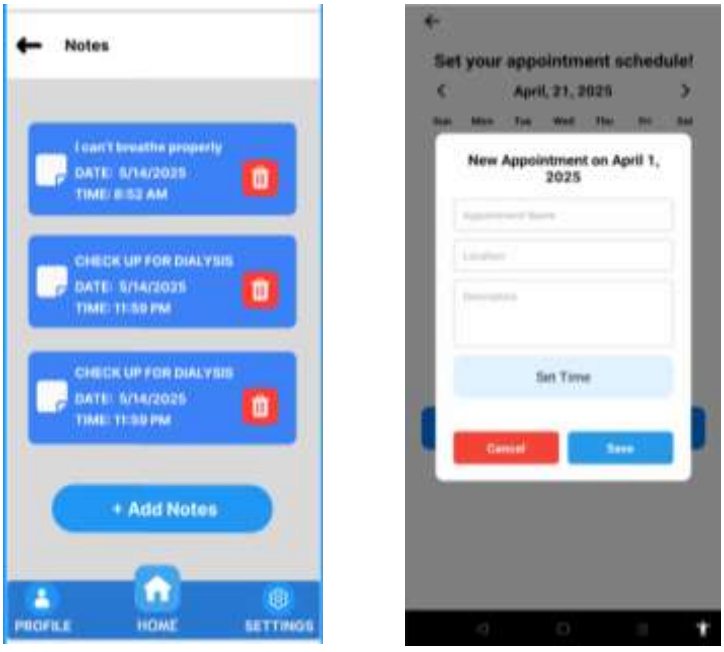
If the user wants to add their medicine reminder, the application shows a "plus" button. It displays a form for the user to input their medication regimen, including medicine name, dose schedule, and reminder frequency. Notifications shows in the notification panel or inside the app. This reminder displays the user's medication details so they can visualize their medication on the app screen.



Settings contains the options that users can configure based on their preferences. The application can also float to the user's screen if allowed, and it has a progress report that shows data analytics on the elderly's medication. Medication history that the dependent can view to monitor and confirm the elderly's medication-taking.

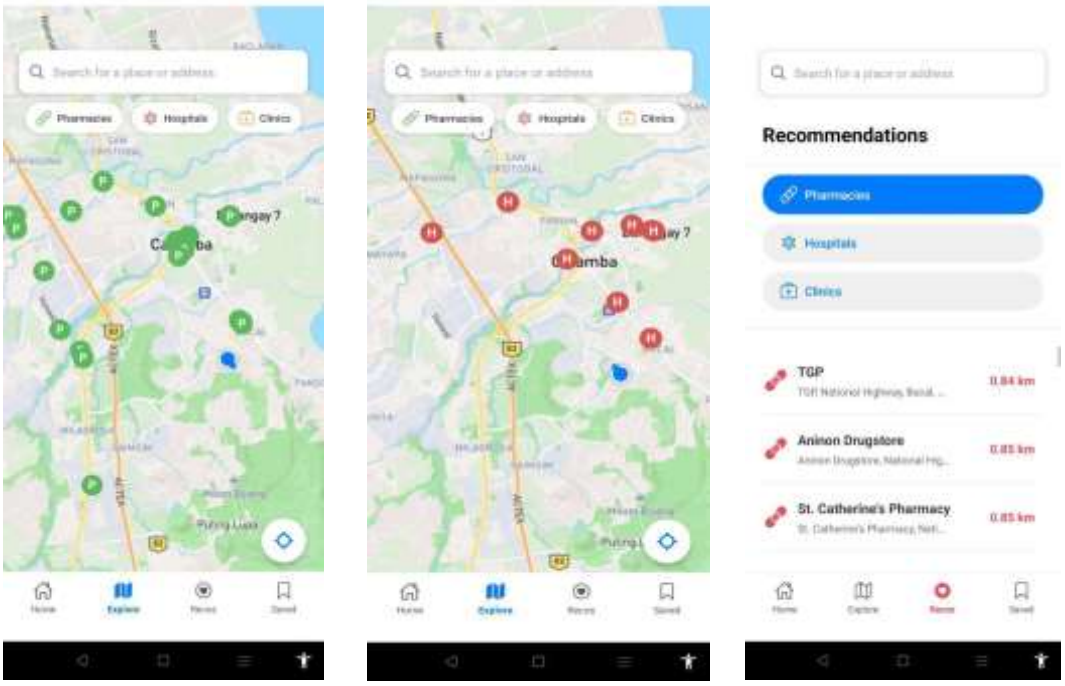


The profile page contains the personal information of the user, it also has a connection code so that the elderly can authorize their caregivers to supervise their medication regimen.



Notes show a button for the user to input and to keep track of their personal health-related information. Appointment Calendar helps users manage their appointments with their doctor and other health-related appointments. Users can

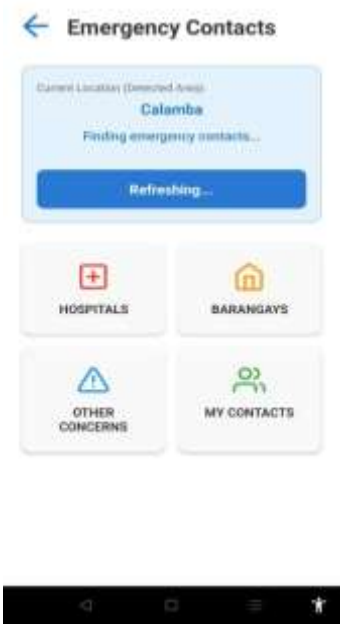
input details such as the appointment and doctor name, location, date and time, and the description of the appointment.



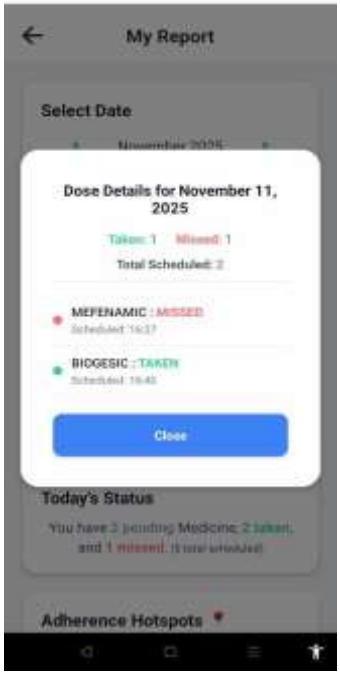
Maps has a mapping feature to help users locate nearby healthcare facilities such as hospitals, clinics, and pharmacies. Users can see their location and search for specific destinations and contact information. The recommendation page suggests the nearest health-related facilities for users to navigate facilities easily.

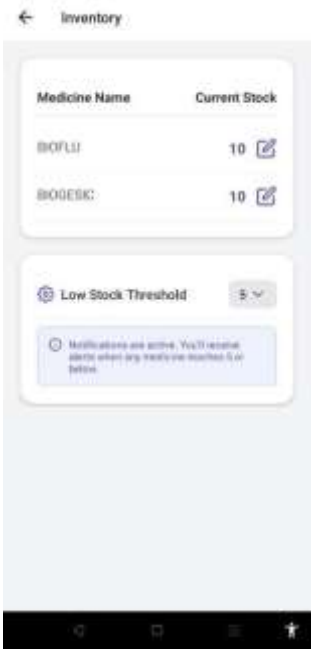
.



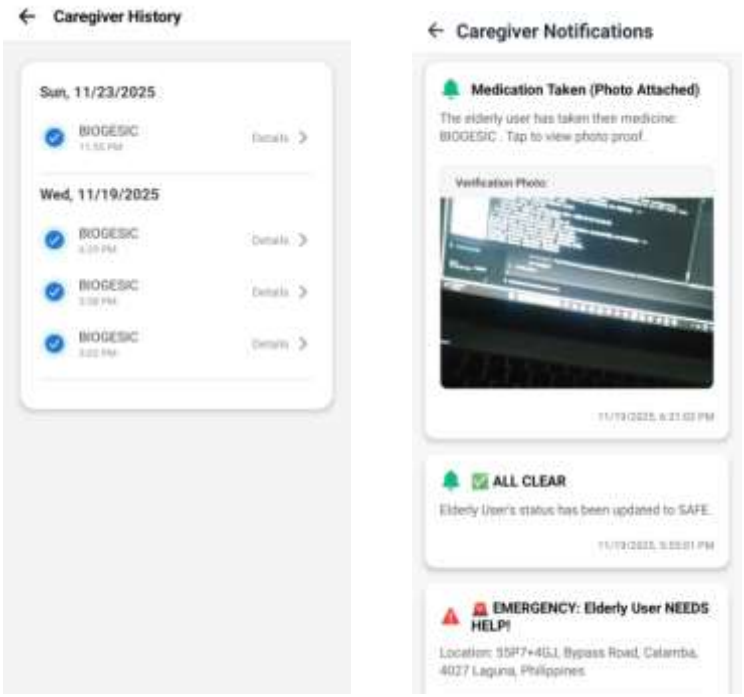


Emergency Contacts are intended to allow quick access to a user's authorized contacts in the situation of a health issue. The app allows users to save the phone numbers and names of their emergency contacts. Emergency Alert is a feature where users can send a notification to their dependents by clicking the “I AM SAFE” button to mark themselves as safe in case of emergency or calamity.





Data analytics provides users with visual information about their medication adherence. This part of the app gathers and evaluates data from the user's activities with the reminder system, including confirmed medicine intakes and missed doses. The findings are shown in charts and graphs, allowing the user to track their adherence as time passes. Inventory can handle the medication stock of the users and send a notification that can notify users about the low stock of the medicine.

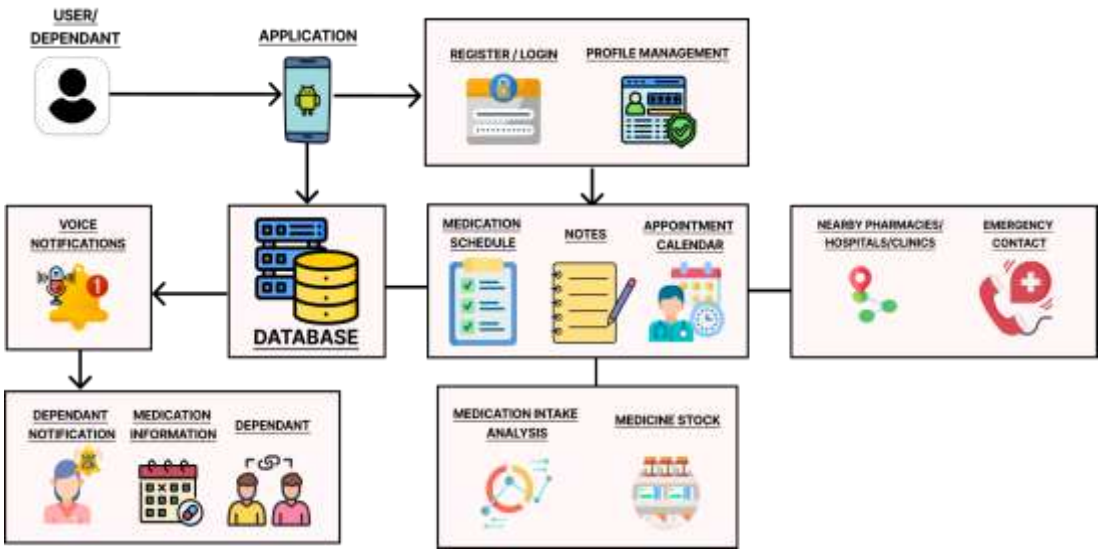


Caregiver history shows the medicines added by the elderly, and notifications serve as proof that elderly individuals take their prescribed medicine and follow its schedule.

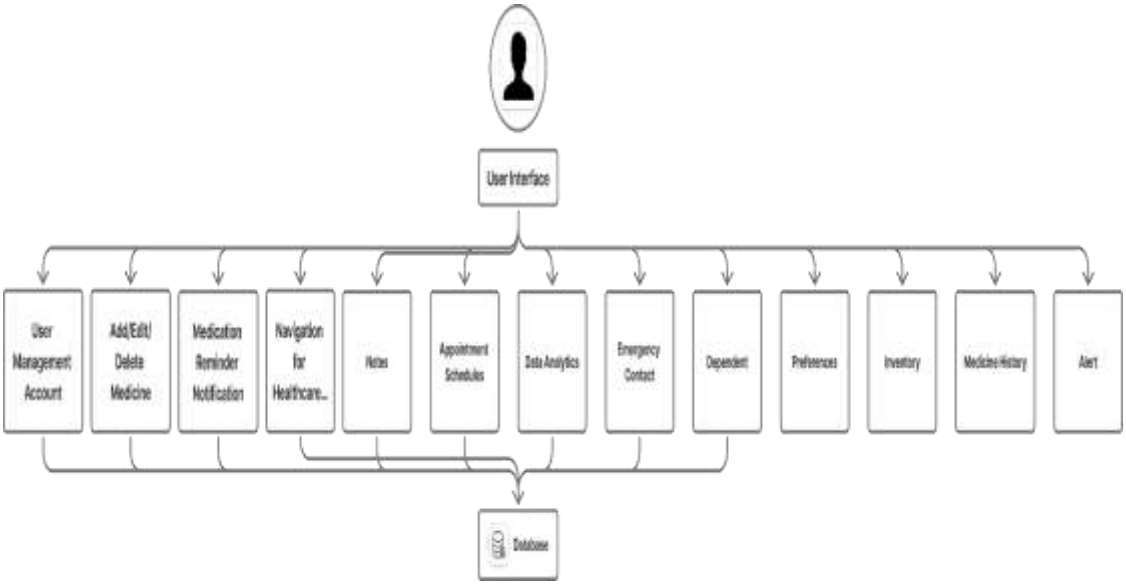
APPENDIX B.

Figures

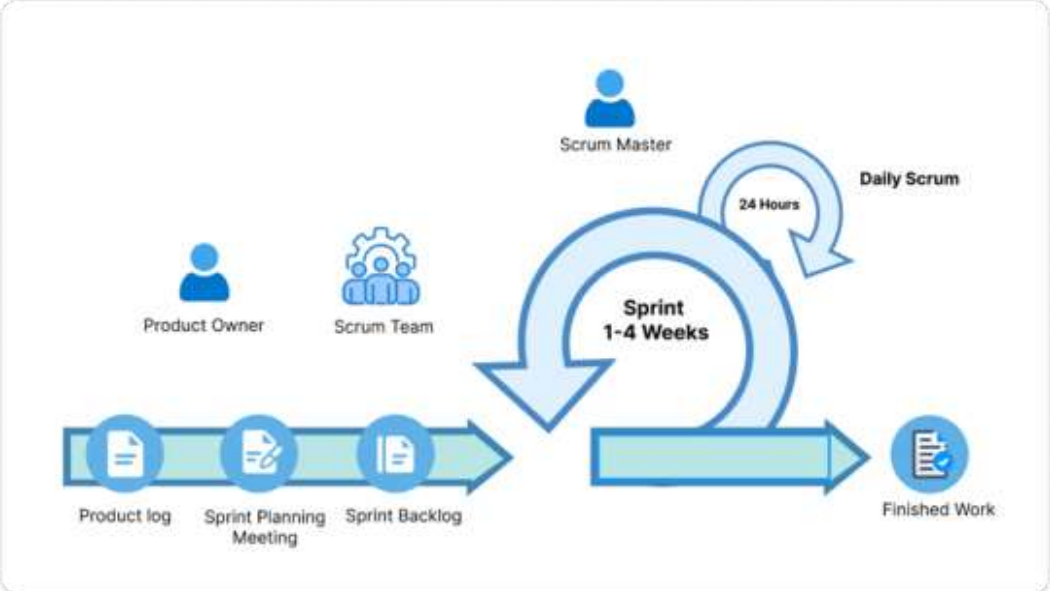
Conceptual Framework



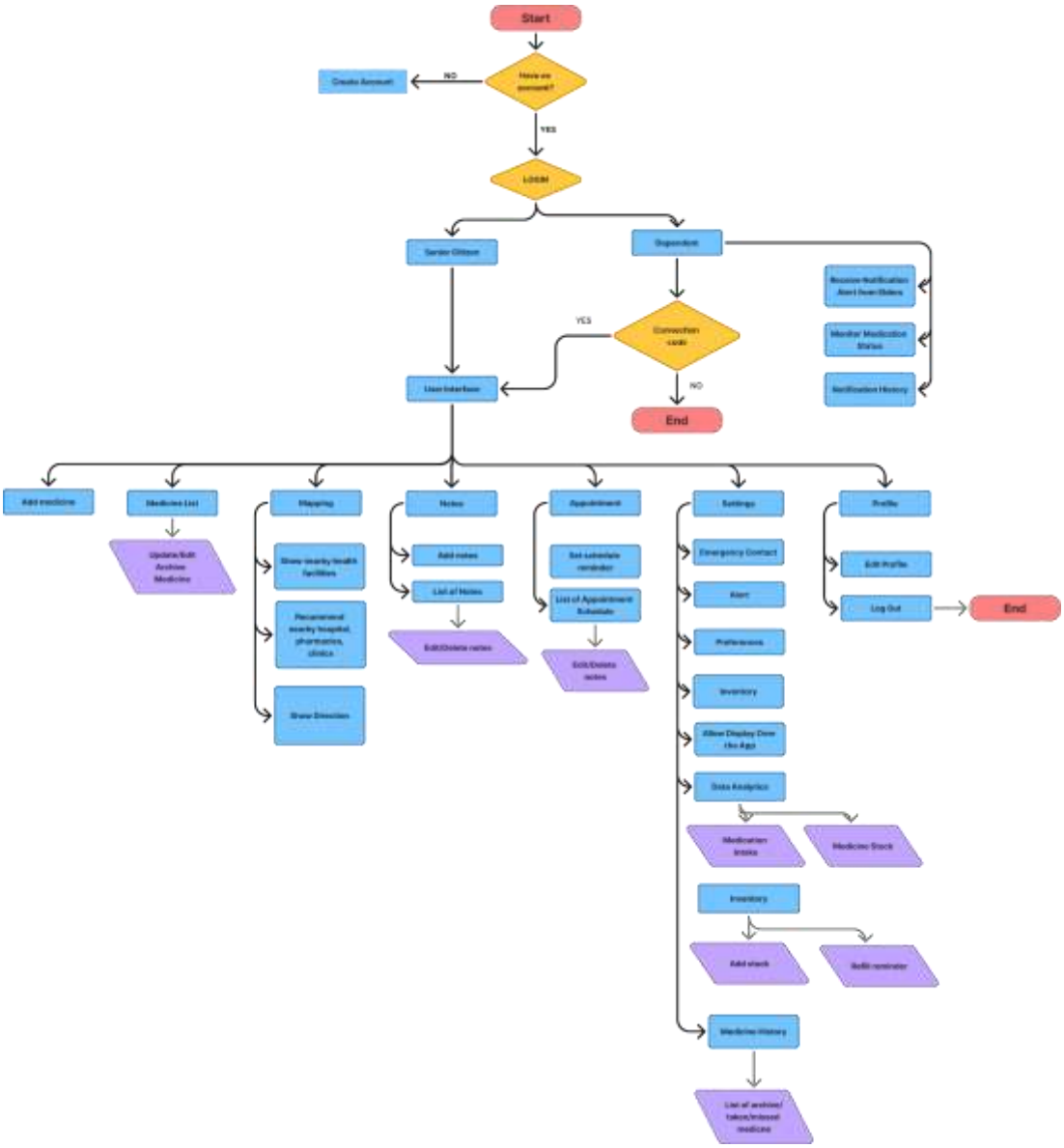
System Architecture



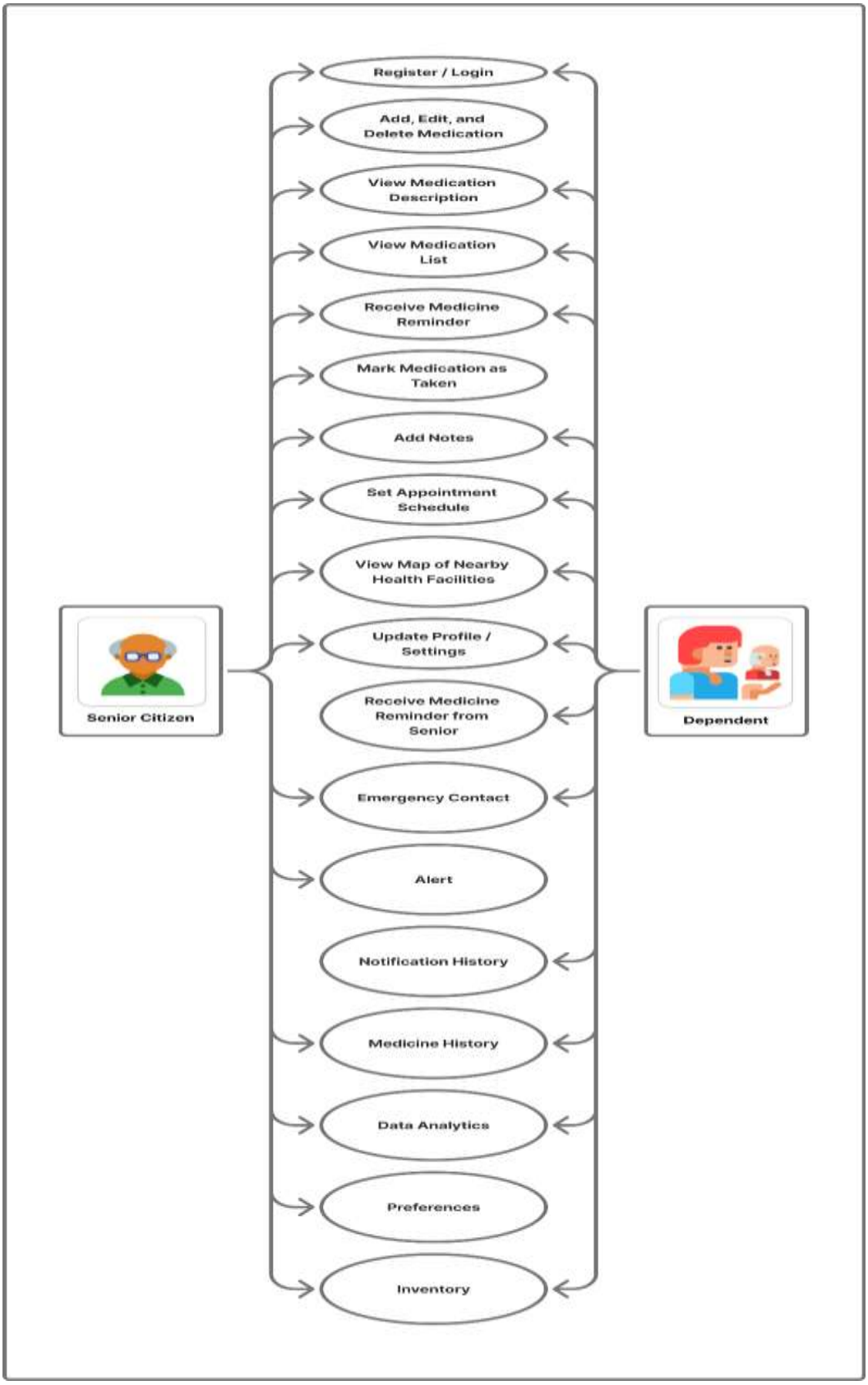
Agile Scrum


Scrum Master
Daily Scrum
24 Hours
Product Owner
Scrum Team
Sprint
1-4 Weeks
Product log
Sprint Planning
Meeting
Sprint Backlog
Finished Work

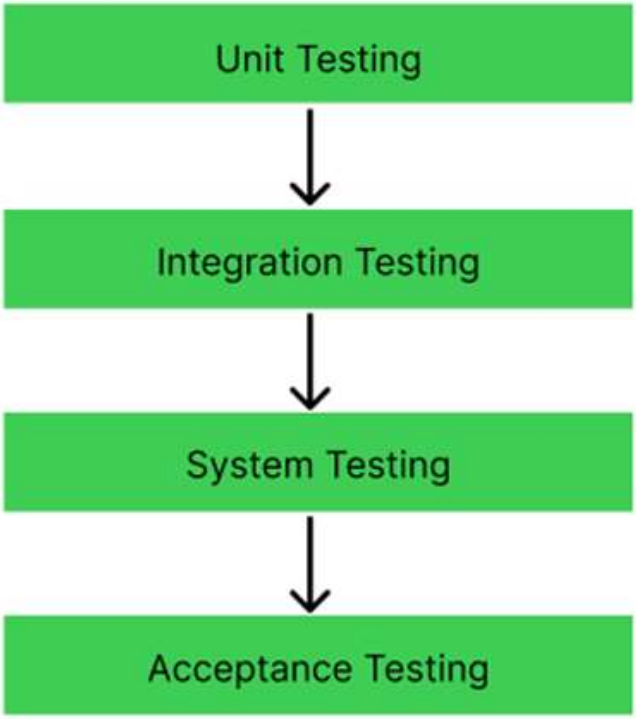
Flowchart



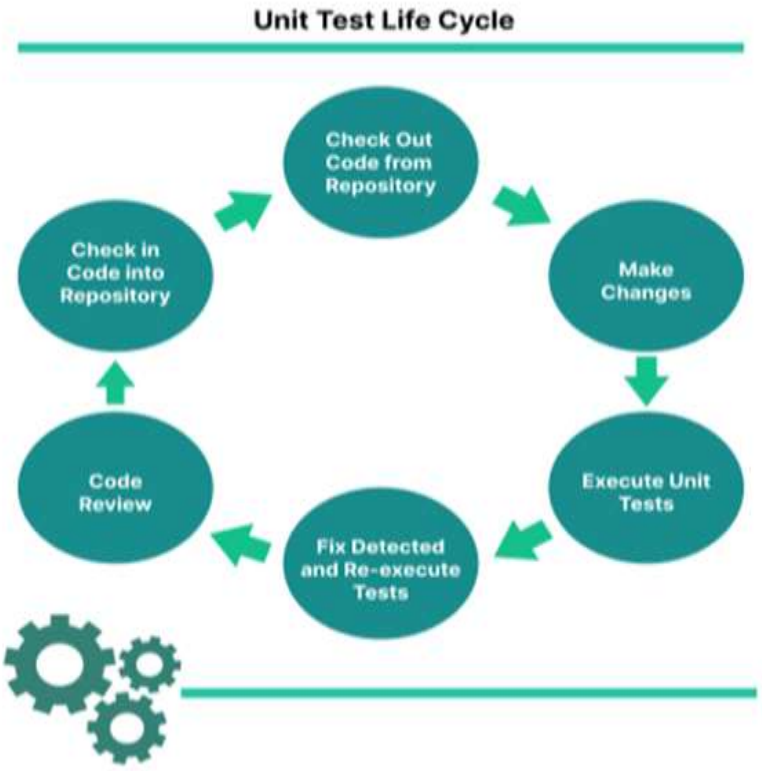
Use Case Diagram



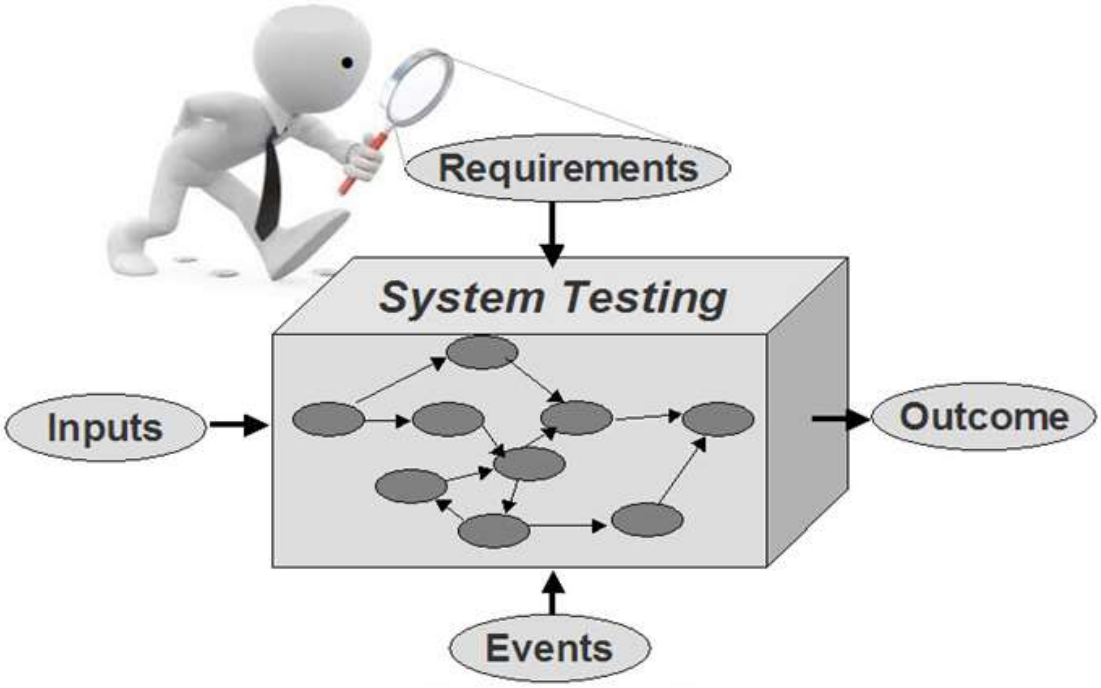
Software Testing



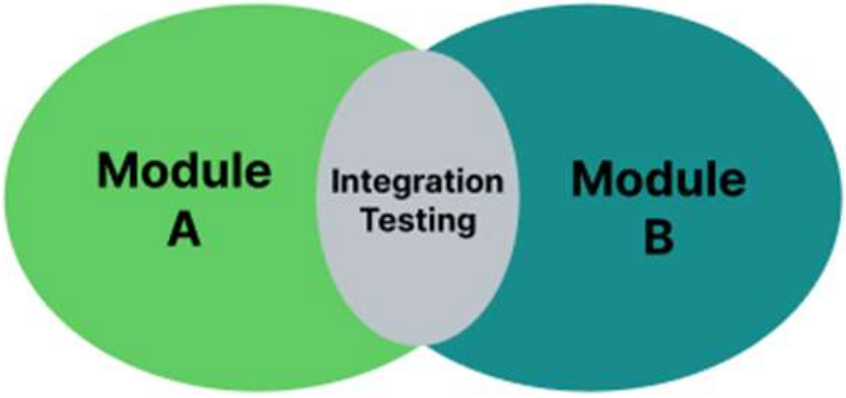
Unit Testing



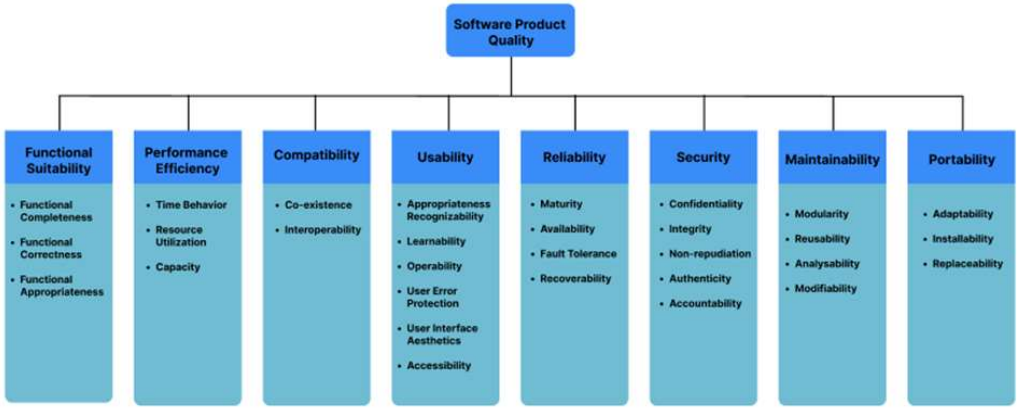
System Testing



Integration testing



ISO 25010 Quality Characteristics



APPENDIX C

Relevant Source Code

```
const HTML_CONTENT = `
<!DOCTYPE html>
<html>
<head>
    <meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=1.0, user-scalable=no">
    <link rel="stylesheet" href="https://unpkg.com/leaflet@1.9.4/dist/leaflet.css" />
    <style>
        body, html { margin: 0; padding: 0; overflow: hidden; width: 100%; height: 100%; }

    #map-container {
    width: 100vw;
    height: 100vh;
    overflow: hidden;
    transform: none !important;
    transition: none !important;
}


        #map { height: 100%; width: 100%; }

        .destination-icon {
            background-color: #E84A5F;
            border-radius: 50%;
            border: 3px solid white;
            width: 20px;
            height: 20px;
            display: flex;
            justify-content: center;
            align-items: center;
            box-shadow: 0 0 5px rgba(0,0,0,0.5);
```

```
        }

        .user-icon {
            position: relative;
            border-radius: 50%;
            border: 2px solid white;
            width: 16px;
            height: 16px;
            background-color: #007AFF;
            display: flex;
            justify-content: center;
            align-items: center;
            box-shadow: 0 0 5px rgba(0,0,0,0.5);
            transform: none;
        }
        .user-icon::after {
            content: '';
            position: absolute;
            top: -7px;
            left: 50%;
            transform: translateX(-50%);
            width: 0;
            height: 0;
            border-left: 5px solid transparent;
            border-right: 5px solid transparent;
            border-bottom: 7px solid #007AFF;
        }

        .custom-div-icon {
            display: flex;
            justify-content: center;
            align-items: center;
```

```
            align-items: center;
        }
    </style>
    <script src="https://unpkg.com/leaflet@1.9.4/dist/leaflet.js"></script>
</head>
<body>
    <div id="map-container">
        <div id="map"></div>
    </div>
    <script>
        const INITIAL_CENTER = [14.17, 121.22];
        const INITIAL_ZOOM = 14;

        const map = L.map('map', {zoomControl: false}).setView(INITIAL_CENTER, INITIAL_ZOOM);
        const mapRotationContainer = document.getElementById('map-container');
        const MAPBOX_ACCESS_TOKEN = 'pk.eyJ1IjoiYWx0aGVhMjEiLCJh1joiV2Imbm1wamw0MDY0bTJqcjM5N2SjOWk4d5J9.YhqpGY2BGSmo3Fmq

        L.tileLayer('https://api.mapbox.com/styles/v1/{id}/tiles/{z}/{x}/{y}?access_token={accessToken}', {
            attribution: '© <a href="https://www.mapbox.com/about/maps/">Mapbox</a> © <a href="http://www.openstreetmap.o
            maxZoom: 18,
            id: 'mapbox/streets-v12',
            tileSize: 512,
            zoomOffset: -1,
            accessToken: MAPBOX_ACCESS_TOKEN
        }).addTo(map);

        let userMarker = null;
        let placeMarker = null;
        let routePolyline = null;
        let poiMarkers = L.layerGroup().addTo(map);
```

```
        let isNavigatingActive = false; // State to track navigation mode

        let fullRoutePoints = [];
        let pastRoutePolyline = null;
        let futureRoutePolyline = null;
        let progressMarker = null;
        window.lastProgressIdx = 0; // Global index to track progress

        const userIconDiv = L.divIcon({
            className: 'user-icon-container',
            html: '<div class="user-icon-rotatable"><div class="user-icon"></div></div>',
            iconSize: [30, 30],
            iconAnchor: [15, 15],
        });

        const destinationIcon = L.divIcon({ className: 'destination-icon', iconSize: [26, 26], html: '<div style="color:w
        const hospitalIcon = L.divIcon({ className: 'custom-div-icon', html: "<div style='background-color:#d9534f;color:
        const pharmacyIcon = L.divIcon({ className: 'custom-div-icon', html: "<div style='background-color:#5cb85c;color:
        const clinicIcon = L.divIcon({ className: 'custom-div-icon', html: "<div style='background-color:#f0ad4e;color:wh

        function getPoiIcon(amenity) {
            if (amenity === 'hospital') return hospitalIcon;
            if (amenity === 'pharmacy') return pharmacyIcon;
            if (amenity === 'clinic') return clinicIcon;
            return hospitalIcon;
        }

        function decodePolyline(encoded) {
            var points = [];
            var index = 0, len = encoded.length;
            var lat = 0, lng = 0;
```

```
        while (index < len) {
            var b, shift = 0, result = 0;
            do {
                b = encoded.charCodeAt(index++) - 63;
                result |= (b & 0x1f) << shift;
                shift += 5;
            } while (b >= 0x20);
            var dlat = ((result & 1) ? ~(result >> 1) : (result >> 1));
            lat += dlat;

            shift = 0;
            result = 0;
            do {
                b = encoded.charCodeAt(index++) - 63;
                result |= (b & 0x1f) << shift;
                shift += 5;
            } while (b >= 0x20);
            var dlng = ((result & 1) ? ~(result >> 1) : (result >> 1));
            lng += dlng;

            points.push([lat / 1e5, lng / 1e5]);
        }

        return points;
    }

    function clearRoute() {
        if (pastRoutePolyline) { map.removeLayer(pastRoutePolyline); pastRoutePolyline = null; }
        if (futureRoutePolyline) { map.removeLayer(futureRoutePolyline); futureRoutePolyline = null; }
        if (progressMarker) { map.removeLayer(progressMarker); progressMarker = null; }
        fullRoutePoints = [];
        window.lastProgressIdx = 0;
    }

    function clearPlaceMarker() { if (placeMarker) { map.removeLayer(placeMarker); placeMarker = null; } }
    function clearPoiMarkers() { poiMarkers.clearLayers(); }

    window.document.addEventListener('message', function(event) {
        const data = JSON.parse(event.data);
        switch (data.type) {
            case 'mapReady':
                window.ReactNativeWebView.postMessage(JSON.stringify({ type: 'mapReady' }));
                break;
            case 'updateLocation':
                if (!userMarker) {
                    userMarker = L.marker([data.lat || INITIAL_CENTER[0], data.lng || INITIAL_CENTER[1]], { icon: use
                    map.setView([data.lat || INITIAL_CENTER[0], data.lng || INITIAL_CENTER[1]], INITIAL_ZOOM);
                } else {
                    userMarker.setLatLng([data.lat, data.lng]);
                }

                if (data.heading !== null && data.heading !== undefined) {
                    const markerElement = userMarker?.getElement();
                    if (markerElement) {
                        const rotatablePart = markerElement.querySelector('.user-icon-rotatable');
                        if (rotatablePart) {
                            rotatablePart.style.transform = 'rotate(' + (data.heading) + 'deg)';
                        }
                    }

                    if (isNavigatingActive) {
                        map.setView([data.lat, data.lng], map.getZoom(), { animate: true, duration: 0.1 });
```

```
                map.setView([data.lat, data.lng], map.getZoom(), { animate: true, duration: 0.1 });
            }
        }

        if (isNavigatingActive && fullRoutePoints.length > 0 && data.lat && data.lng) {
            const userLatLng = L.latLng(data.lat, data.lng);
            const SNAP_DISTANCE = 50;
            let closestIdx = 0;
            let minDist = Infinity;

            for (let i = window.lastProgressIdx; i < fullRoutePoints.length; i++) {
                const point = L.latLng(fullRoutePoints[i][0], fullRoutePoints[i][1]);
                const d = map.distance(userLatLng, point);
                if (d < minDist) {
                    minDist = d;
                    closestIdx = i;
                }
            }

            if (
                closestIdx > window.lastProgressIdx &&
                (closestIdx - window.lastProgressIdx) < 50
            ) {
                window.lastProgressIdx = closestIdx;
                window.ReactNativeWebView.postMessage(
                    JSON.stringify({
                        type: 'updateProgress',
                        payload: { progressIdx: window.lastProgressIdx },
                    })
                );
            }
```

```
                break;
            case 'clearMap':
                clearPlaceMarker();
                clearRoute();
                clearPoiMarkers();
                mapRotationContainer.style.transform = 'rotate(0deg) scale(1)';
                const markerElement = userMarker?.getElement();
                if(markerElement) {
                    const rotatablePart = markerElement.querySelector('.user-icon-rotatable');
                    if (rotatablePart) {
                        rotatablePart.style.transform = 'rotate(0deg)';
                    }
                }
                break;
            case 'recenterMap':
                map.setView([data.lat, data.lng], map.getZoom(), { animate: true, duration: 1 });
                break;

                case 'rotateMapTo':
if (isNavigatingActive) {
    const angle = data.angle || 0;
    mapRotationContainer.style.transform =
        'rotate(' + (-angle) + 'deg)';
    if (userMarker) {
        const markerElement = userMarker.getElement();
        if (markerElement) {
            const rotatablePart = markerElement.querySelector('.user-icon-rotatable');
            if (rotatablePart) {
                rotatablePart.style.transform = 'rotate(' + angle + 'deg)';
            }
        }
    }
```

```
            mapRotationContainer.style.transform = 'none';
            const markerElement = userMarker?.getElement();
            if (markerElement) {
                const rotatablePart = markerElement.querySelector('.user-icon-rotatable');
                if (rotatablePart) {
                    rotatablePart.style.transform = 'rotate(0deg)';
                }
            }
        }
        break;
    case 'showPlace':
        clearPlaceMarker();
        clearPoiMarkers();
        placeMarker = L.marker([data.lat, data.lng], { icon: destinationIcon }).bindPopup(data.name || 'Destination').addT
        map.setView([data.lat, data.lng], 16);
        break;
    case 'showRoute':
        clearRoute();
        clearPlaceMarker();
        clearPoiMarkers();

        if (!data.polyline) {
            window.ReactNativeWebView.postMessage(JSON.stringify({ type: 'routingError', payload: { message: 'Polyline dat
            return;
        }

        fullRoutePoints = decodePolyline(data.polyline);

        if (data.startLat && data.startLng) {
            const firstPoint = fullRoutePoints[0];
            const startDist = L.latLng(firstPoint[0], firstPoint[1]).distanceTo(L.latLng(data.startLat, data.startLng));
```

```
        }
    }

    window.lastProgressIdx = 0;
    futureRoutePolyline = L.polyline(fullRoutePoints, { color: '#007AFF', weight: 6, opacity: 0.7 }).addTo(map);

    placeMarker = L.marker(fullRoutePoints[fullRoutePoints.length - 1], { icon: destinationIcon })
        .bindPopup(data.destinationName || 'Destination')
        .addTo(map)
        .on('click', () => window.ReactNativeWebView.postMessage(JSON.stringify({ type: 'markerClick' })));

    map.fitBounds(futureRoutePolyline.getBounds(), { padding: [50, 50] });
    window.ReactNativeWebView.postMessage(JSON.stringify({ type: 'routeDrawn' }));
    break;
case 'showPois':
    clearPlaceMarker();
    clearRoute();
    clearPoiMarkers();
    if (data.pois && data.pois.length > 0) {
        data.pois.forEach(poi => {
            const marker = L.marker([poi.latitude, poi.longitude], { icon: getPoiIcon(poi.amenity) })
                .bindPopup(poi.name)
                .on('click', () => {
                    window.ReactNativeWebView.postMessage(JSON.stringify({ type: 'poiClick', payload: poi }));
                });
            poiMarkers.addLayer(marker);
        });
        map.fitBounds(poiMarkers.getBounds(), { padding: [50, 50] });
    }
    break;
case 'clearMap':
    clearPlaceMarker();
```

APPENDIX D

Evaluation Tool

ISO 25010 QUESTIONNARE

Dear Respondent,

We are fourth-year students currently pursuing the **Bachelor of Science in Information Technology**, specializing in **Web and Mobile Application Development**. As part of the academic requirements, we have developed an **Android-Based Medication Reminder Application Integrating GIS-Based Mapping for Elderly Individuals**, which adheres to the **ISO/IEC 25010 Software Quality Model.**

To evaluate the system's effectiveness and overall quality, we are conducting an assessment focusing on key aspects such as **usability, functionality, security, and user experience.**

Your participation in this evaluation is highly appreciated. Kindly assess each item based on your experience with the system, using the **4-point rating scale** provided below. Your responses are treated with the utmost confidentiality and used solely for academic purposes.

Thank you for your time and cooperation.

Direction: Encircle the rubric that corresponds to your evaluation

Category	Subcategory	1–Strongly Disagree	2 – Disagree	3 – Agree	4 – Strongly Agree
1. Functional Suitability	**Functional Completeness**	Major functions missing; fails to meet requirements.	Most core functions present but some missing or incomplete.	All required functions implemented with minor gaps.	Fully complete; all required functionalities implemented correctly.
	Functional Correctness	Frequent errors and incorrect outputs.	Some errors occur; limited accuracy.	Outputs mostly correct; minor inaccuracies.	Completely accurate and reliable outputs.
	Functional Appropriateness	Functions poorly aligned to user goals; inefficient workflow.	Some functions helpful but not optimal.	Functions generally useful and appropriate.	Functions fully align with user goals; highly efficient workflow.
2. Performance Efficiency	**Time Behavior**	Very slow; significant lag or delays.	Noticeable performance issues.	Responds in reasonable time under typical load.	Fast, responsive even under heavy load.
	Resource Utilization	Consumes excessive CPU/memory.	Acceptable but occasionally inefficient.	Uses resources efficiently most of the time.	Highly optimized; minimal resource usage.
	Capacity	Fails with large data or users.	Handles moderate load; struggles at scale.	Works well with expected load.	Scales efficiently to handle high capacity.
3. Compatibility	**Co-existence**	Conflicts with other systems or software.	Limited compatibility; requires adjustments.	Coexists well with related systems.	Fully interoperable without conflict.

	Interoperability	Cannot exchange data with other systems.	Basic integration possible with effort.	Integrates with some external systems smoothly.	Seamless data exchange with multiple systems.
4. Usability	**Appropriateness Recognizability**	Users cannot easily recognize purpose.	Purpose somewhat unclear; moderate confusion.	Purpose understandable with minimal training.	Instantly recognizable and intuitive.
	Learnability	Difficult to learn; poor guidance.	Steep learning curve; limited support.	Easy to learn with brief orientation.	Instantly learnable; intuitive without instructions.
	Operability	Complicated operations; many errors.	Somewhat usable but unintuitive.	Easy to operate with minor usability issues.	Smooth, intuitive operation throughout.
	User Error Protection	Frequent user errors; poor feedback.	Some validation but limited prevention.	Adequate error prevention and feedback.	Strong safeguards; clear recovery and prevention.
	User Interface Aesthetics	Unappealing and inconsistent.	Acceptable visuals; lacks consistency.	Visually pleasing and mostly consistent.	Polished, professional, and consistent design.
	Accessibility	Inaccessible to users with disabilities.	Limited accessibility features.	Basic accessibility provided.	Fully accessible; meets accessibility standards.
5. Reliability	**Maturity**	Frequent crashes or failures.	Occasional failures under stress.	Stable with rare failures.	Highly stable and reliable under all conditions.
	Availability	Often unavailable or downtime-prone.	Sometimes inaccessible.	Generally available; brief downtimes.	Almost always available; minimal downtime.
	Fault Tolerance	Crashes on error; no recovery.	Partially recovers from faults.	Recovers gracefully from most errors.	Fully fault-tolerant with redundancy and auto-recovery.
	Recoverability	Data loss common after failure.	Partial recovery possible.	Full recovery possible with effort.	Automatic and complete recovery after failure.
6. Security	**Confidentiality**	No protection of sensitive data.	Basic security; potential leaks.	Data secured with standard measures.	Strong encryption and strict confidentiality controls.
	Integrity	Data easily altered or corrupted.	Limited data validation.	Protects data integrity with checks.	Strong validation; prevents all unauthorized changes.
	Non-repudiation	No record of user actions.	Partial audit logs; incomplete tracking.	Logs all major actions.	Complete audit trail preventing repudiation.

	Accountability	No user accountability.	Basic authentication only.	Users identifiable and accountable.	Full accountability with audit and role-based control.
	Authenticity	Weak or missing authentication.	Simple credentials; easily compromised.	Secure authentication mechanisms in place.	Multi-factor authentication and strong identity management.
7. Maintainability	**Modularity**	Code tightly coupled; hard to modify.	Some modularity; changes affect multiple parts.	Reasonably modular and organized.	Highly modular; components easily replaced or updated.
	Reusability	Code cannot be reused.	Some reusable parts but limited.	Moderate reusability across modules.	Highly reusable, well-documented components.
	Analyzability	Difficult to diagnose issues.	Basic diagnostics possible.	Issues traceable with some effort.	Easy to analyze and debug with clear logs.
	Modifiability	Changes often break functionality.	Edits possible but risky.	Modifiable with minimal side effects.	Highly adaptable and stable when modified.
	Testability	Hard to test; lacks structure.	Testing possible but limited automation.	Structured and testable with tools.	Fully testable; automated and well-covered.
8. Portability	**Adaptability**	Works only in one environment.	Limited adaptability to other environments.	Runs across multiple platforms with minor tweaks.	Fully adaptable and environment-agnostic.
	Installability	Installation is complex and error-prone.	Installable with effort or support.	Straightforward installation process.	Simple, seamless installation and setup.
	Replaceability	Cannot replace other systems.	Replacement possible but disruptive.	Can replace existing system with minimal impact.	Easily replaces or integrates with other systems.

Validation of expert:

_____ _____

Signature over printed name Date

_____

Designation/Position

_____

Office

Technology Acceptance Model(TAM)

Med-Alalay: **Android-Based Medication Reminder Application Integrating GIS-Based Mapping for Elderly Individuals**

Statements	(Strongly Disagree / Lubos na Hindi Sumasang-ayon) = 1	(Disagree / Hindi Sumasang-ayon) = 2	(Agree / Sumasang-ayon) = 3	(Strongly Agree / Lubos na Hindi Sumasang-ayon) = 4
A. Perceived Usefulness				
1. The application helps me remember the time to take my medicine.	Does not help at all in reminding medicine time.	Helps a little but often fails to remind. .	Usually helps remind me to take medicine.	Always helps and effectively reminds me of medicine time.
2. The voice notification is useful for medication reminders.	Voice notification is not useful or unclear.	Sometimes useful but not consistent.	Usually useful and understanda ble.	Very useful and clear in reminding me about medicine.
3. The map helps me quickly find the nearest hospital or pharmacy.	Does not help me locate facilities.	Helps but takes time or is confusing.	Helps me find facilities most of the time. .	Quickly and easily helps me locate the nearest facilities.
4. The Health Notes helps me record important information about my health.	Cannot record or manage health notes easily.	Records some information but not complete. .	Helps record most health details clearly.	Very useful for recording and organizing health information.
5. Using the application contributes to better management of my health.	Does not improve my health management.	Slightly improves how I manage health.	Improves my health care managemen t.	Greatly improves and supports my health management.
B. Perceived Ease of Use				

1. The application is easy to use and understand.	Very difficult to use or confusing.	Somewhat easy but needs assistance.	Generally easy to use and understand.	Very easy and intuitive to use.
2. The voice notification is clear and easy to understand.	Unclear and confusing notifications.	Sometimes clear but inconsistent.	Mostly clear and easy to follow.	Always clear and easily understood.
3. I can easily set the schedule for taking medicine.	Very hard to set reminders.	Can set reminders but takes time.	Easy to set reminders with little effort.	Very easy and quick to set reminders.
4. The design of the application is orderly and simple. .	Design is cluttered or confusing.	Design is acceptable but needs improvement.	Design is organized and easy to follow.	Design is very simple, clear, and senior-friendly.
5. I can easily find and use the map feature. .	Difficult to find or use the map feature.	Works but not easy to access.	Generally easy to use the map.	Very easy and helpful to use.
C. Attitude				
1. I enjoy using the App.	I do not enjoy using it.	I sometimes enjoy it but with issues.	I enjoy using it most of the time.	I greatly enjoy and feel happy using it.
2. I feel confident while using the application.	I feel unsure or nervous using it.	I am somewhat confident.	I feel confident using it.	I feel very confident and comfortable.
3. Using the application helps me become more responsible.	Does not help me be responsible.	Slightly improves my responsibility.	Helps me remember to take medicine.	Greatly helps me be consistent and responsible.
4. The application makes managing my health easier.	Does not make managing health easier.	Somewhat helps manage health.	Makes health managemen t easier. .	Greatly simplifies my health management.

5. Having all features in one app is a great idea.	I don't think it's a good idea.	The idea is acceptable but not perfect.	The idea is good and practical.	Excellent idea and very useful.
D. Behavior				
1. I intend to continue using the application daily. .	I don't plan to use it.	I might use it sometimes.	I plan to use it regularly. .	I definitely use it daily.
2. I would recommend this application to others.	I won't recommend it.	Might recommend with hesitation.	I recommend it.	Strongly recommend to others.
3. I rely on this app for medication reminders.	I don't trust it as reminder. .	I'll rely sometimes.	I'll rely on it often.	I'll fully rely on it.
4. I plan to use the app's various features daily.	I won't use its features.	I'll try some features.	I'll use main features often.	I'll use all features daily.
5. I rely on this system to help maintain my good health.	I won't rely on it for health.	I might use it sometimes.	I'll rely on it for health reminders.	I'll fully depend on it for maintaining health.

Validated by:

_____

APPENDIX E.

Publishable Paper

MED-ALALAY: ANDROID-BASED MEDICATION REMINDER APPLICATION INTEGRATING GIS-BASED MAPPING FOR ELDERLY INDIVIDUALS

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Article History

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Medication Adherence
Elderly Care
Voice Notification
GIS-Mapping
Mobile Health Application
Android Development

ABSTRACT

Medication non-adherence among elderly individuals remains a critical healthcare challenge, leading to increased hospitalizations and mortality rates. This study developed MED-ALALAY, an Android-based medication reminder application integrating voice notifications and Geographic Information System (GIS) mapping to assist elderly individuals in Los Baños, Laguna with prescribed medication intake. The application was developed using Agile Scrum methodology with React Native and TypeScript, incorporating features including voice-based reminders, dependent monitoring, health notes, appointment scheduling, and navigation to nearby healthcare facilities. Evaluation using the Technology Acceptance Model (TAM) with 50 users and ISO 25010 standards with 3 IT professionals results demonstrated that voice notifications significantly improved medication adherence, while GIS integration enhanced accessibility to healthcare resources. The dependent module successfully enabled real-time monitoring by caregivers. The system proved effective in addressing medication management challenges, providing an accessible and user-friendly solution for elderly individuals requiring assistance with complex medication regimens.

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I. INTRODUCTION

Medicine is an essential component of health, as it is a primary tool for managing, preventing, and treating illnesses. Positive health results are achievable if medication is followed as prescribed by the doctors, the efficiency of the drugs depends on how the medicine is correctly taken, which is why adherence to medication is important to improve quality of life and life expectancy. Non-adherence persists as a major problem in the world today and has been associated with higher mortality, worsening of medical conditions, and higher hospitalization rates, especially among elderly individuals.

One of the essential parts in achieving positive health outcomes is

medication adherence. It is based on what their doctors give as the adherence of the patients when taking their prescribed medicine and schedule. It is not only effective only by the medicine formula but also if it is managed correctly by the prescribed schedule. Non-adherence is considered a major problem, as it results in higher hospitalization rates, higher mortality, and worsening of medical conditions, especially among elderly individuals.

Elderly individuals are the ones who have the most challenges because they usually deal with a variety of medical problems that require complicated drug regimens. It might be difficult for older persons to take their medications on their own due to cognitive decline, vision impairment, physical vulnerability, and neglect. In the Philippines, many elderly individuals rely on their memory. It shows the importance of having a dependable and effective way to assist elderly individuals in adhering to their prescribed schedules. It is more possible that elders may forget or manage their dosages incorrectly. This approach encourages the researcher to assist in promoting medication adherence, specially intended to support elderly individuals in effectively complying with their prescription schedules.

To address this issue, the researchers propose the idea of developing an Android-based medicine reminder application for elderly individuals. The app features medicine notifications about the prescribed schedule of their medication that notifies them to take their medicine. Furthermore, there's a dependent feature where family members or caregivers be notified about the medication time. Creating an accessible and easy-to-use reminder application aims to support elderly individuals and their families. The study was a step toward helping elderly individuals to live independently and safely, in addition to being a reliable option. It can help not only elderly individuals and families but also the health sector to address the issue and focus on the concerns about medication adherence for elderly individuals.

Project Objective

This project aims to develop an Android-based medicine reminder for the elderly individuals. Therefore, Voice Notification was integrated to improve medication adherence. Furthermore, the project specifically aims to:

1. To develop an Android-based medicine reminder application that assists the elderly in prescribed medication intake using voice notification.
2. To incorporate a navigation feature that provides the location of nearby healthcare facilities for elderly individuals.
3. Evaluate the developed project to substantiate the accessibility and quality of the mobile application based on the ISO 25010 software standard

II. METHODOLOGY

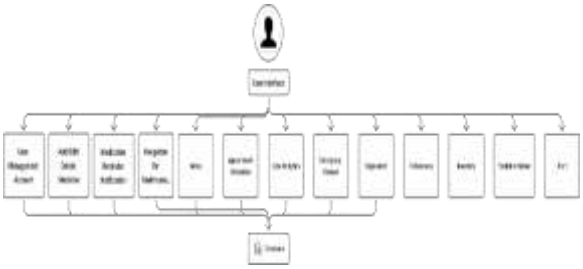
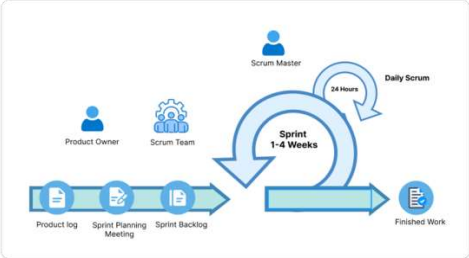
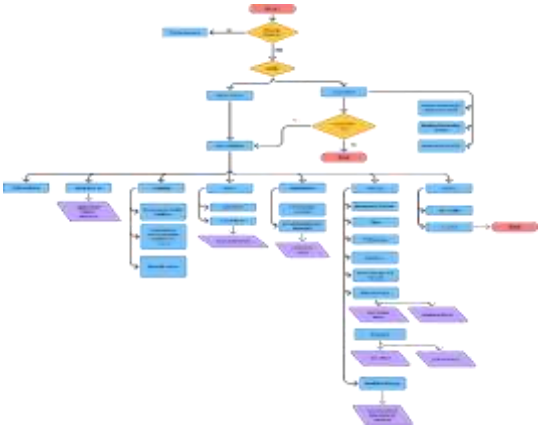


Figure 1 illustrates the architecture of the MED-ALALAY system. The user interface serves as the main access point to all functionality. Users can register and log in to create personal accounts and access their information. The medication management module allows users to add, edit, and delete medicines from their list as needed. The reminder notification feature delivers alerts through voice notifications, helping users take their prescribed medicines as scheduled to ensure consistent medication consumption. /The system includes a navigation module that assists users in locating nearby hospitals, pharmacies, and clinics in their area. A notes feature acts as a recording function that stores users' important health information. Additionally, users can view and monitor their medical appointments through the appointment scheduling feature. If a user skips or misses a dose, the dependent module notifies family members or caregivers. A centralized Firebase database stores all data produced by these features, ensuring data accuracy and security throughout the system.

Project Design

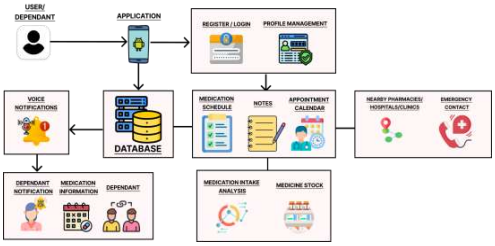


The development process followed the Agile Scrum methodology, enabling iterative development and continuous feedback incorporation. The research was conducted in Los Baños, Laguna, selected for its diverse population of elderly individuals and proximity to various healthcare facilities. The project utilized purposive sampling to identify participants who would benefit most from the application.



Every sprint began with a sprint planning meeting where the team selected the most essential features from the product backlog and worked for one to four weeks to design and test them. Daily scrum meetings helped track progress, address blockers, and align efforts. At the end of each sprint, a functional and testable version of the system was produced, ready for feedback and evaluation by elderly users and their caregivers to ensure the app's efficiency and usability.

B. System Architecture and Development



The application was developed using React Native and TypeScript for cross-platform compatibility, with Firebase Realtime Database for secure data storage. The system architecture comprises eight integrated modules: Information Module (medication details, notes, calendar), GIS Mapping Module (location services via Mapbox API and OpenStreetMap), Notification Module (text-to-speech technology for voice alerts), Data Analytics Module (adherence tracking and visualization), Dependent Module (caregiver monitoring), Database Module (centralized data repository), User Account Management Module (authentication and security), and Emergency Contact Module (hotlines and emergency contacts).

Voice notifications are implemented using text-to-speech (TTS) technology to convert medication information into audible alerts. The GIS mapping feature integrates Mapbox API to display nearby pharmacies, hospitals, and clinics with real-time navigation capabilities. The dependent module enables authorized caregivers to receive notifications when elderly users miss scheduled doses.



C.Testing and Evaluation

The system underwent comprehensive testing including unit testing (individual module functionality), integration testing (inter-module communication), system

testing (overall performance), and acceptance testing (user satisfaction). Software quality was evaluated using ISO 25010 standards assessing functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability.

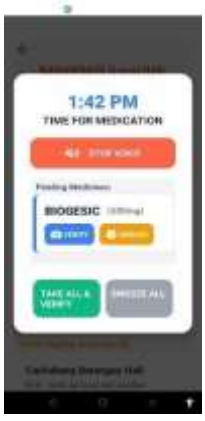
D. Population and Data Collection

CATEGORY OF RESPONDENTS	SAMPLE
User	50
IT Professional	3
Total	53

The study involved 53 respondents: 50 end-users (elderly individuals, family members, and caregivers) and 3 IT professionals. Data collection utilized structured questionnaires administered via hard copy surveys. The Technology Acceptance Model (TAM) framework assessed perceived usefulness, perceived ease of use, attitude, and behavioral intentions using a 4-point Likert scale. Statistical analysis employed mean scores and standard deviations with the following interpretation: 3.25-4.00 (Strongly Agree), 2.50-3.24 (Agree), 1.75-2.49 (Disagree), and 1.0-1.74 (Strongly Disagree).

III. RESULTS

System Development and Features





The MED-ALALAY application was successfully developed with all planned modules functioning correctly. The voice notification feature demonstrated clear audibility with medication names and schedules announced at prescribed times. The GIS mapping module accurately displayed nearby healthcare facilities within Los Baños, Laguna, with functional navigation capabilities. Integration testing confirmed seamless data flow between modules with average processing times of 2-5 seconds per operation.

B. User Acceptance (TAM Evaluation)

Item	Mean	SD	Interpretation
A. Perceived Usefulness	3.17	0.85	Agree
B. Perceived Ease of Use	3.24	0.77	Agree
C. Attitude	3.27	0.79	Strongly Agree
D. Behavioral Intention	3.08	0.83	Agree
OVERALL WEIGHTED MEAN	**3.19**	**0.81**	**Agree**

The Technology Acceptance Model evaluation yielded an overall weighted mean of 3.19 (SD=0.81), interpreted as "Agree." Perceived usefulness achieved a mean of 3.17 (SD=0.85), with the medication reminder function receiving the highest rating (3.48, Strongly Agree) recognizing the application as useful tool. Perceived ease of use scored 3.24 (SD=0.77), with design simplicity and layout rated at 3.46 (Strongly Agree). Attitude toward the system obtained 3.27 (SD=0.79, Strongly Agree), with respondents particularly appreciating the integration of multiple health management

features (3.42). Behavioral intention scored 3.08 (SD=0.83, Agree), with 3.60 (Strongly Agree) for willingness to recommend the application to others.

C. Software Quality (ISO 25010 Evaluation) Test and Evaluate the

Category	Mean	SD	Interpretation
Functional Suitability	3.78	0.39	Strongly Agree
Performance Efficiency	3.89	0.19	Strongly Agree
Compatibility	3.50	0.58	Strongly Agree
Usability	3.72	0.48	Strongly Agree
Reliability	3.50	0.29	Strongly Agree
Security	3.73	0.35	Strongly Agree
Maintainability	3.87	0.23	Strongly Agree
Portability	3.89	0.19	Strongly Agree
Composite	**3.74**	**0.34**	**Strongly Agree**

Software using ISO 25010

ISO 25010 assessment by IT professionals yielded a composite score of 3.74 (SD=0.337), interpreted as "Strongly Agree." Individual category results included: Functional Suitability (3.78), Performance Efficiency (3.89), Compatibility (3.50), Usability (3.72), Reliability (3.50), Security (3.73), Maintainability (3.87), and Portability (3.89). These scores confirm that the system meets international software quality standards across all evaluated dimensions.

D. System Testing Results

All testing phases are successfully completed. Unit testing verified correct functionality of individual modules with no critical errors. Integration testing confirmed proper inter-module communication with data consistency maintained across operations. System testing demonstrated stable performance under various usage scenarios. Acceptance testing indicated high user satisfaction with practical applicability in real-world medication management scenarios.

IV. DISCUSSION

The results demonstrate that MED-ALALAY effectively addresses medication non-adherence challenges among elderly individuals through integrated voice notifications and GIS mapping. The high user acceptance ratings (TAM overall mean: 3.19) indicate that the application successfully meets user needs while maintaining ease of use. The voice notification feature proved particularly valuable, as evidenced by the excellent rating for medication reminders (3.48), addressing visual impairment and memory-related challenges common among elderly users.

The GIS mapping integration received positive feedback (3.16, Agree), providing practical utility for locating healthcare resources. This feature addresses a critical gap in medication management by ensuring elderly users can easily access pharmacies for prescription refills and medical facilities for consultations. The dependent monitoring module strengthened caregiver involvement, enabling remote support without requiring constant physical presence—a significant advantage for families with busy schedules.

ISO 25010 evaluation results (composite: 3.74) confirm the application's technical robustness. High scores in performance efficiency (3.89) and portability (3.89) indicate the system operates smoothly across various Android devices. Strong security (3.73) and reliability (3.50) ratings ensure user data protection and consistent functionality. The excellent maintainability score (3.87) suggests the system can be efficiently updated and improved based on user feedback.

The behavioral intention results (3.08, Agree) with high recommendation willingness (3.60, Strongly Agree) suggest strong potential for adoption and sustainability. Users' positive attitudes (3.27, Strongly Agree) toward the integrated feature set indicate that combining multiple health management tools within a single application enhances perceived value and utility.

Study limitations include geographic restriction to Los Baños, Laguna, and the relatively small sample size. The voice notification system's pronunciation accuracy is limited by available language options. Future enhancements should include expanded mapping coverage, improved voice recognition for complex medication names, multilingual support, and integration with electronic health records for automated prescription updates.

V.CONCLUSIONS

MED-ALALAY successfully addresses medication adherence challenges among elderly individuals through voice-enabled reminders and GIS-based navigation to healthcare facilities. The application achieved excellent ratings in both user acceptance (TAM: 3.19/4.0) and software quality standards (ISO 25010: 3.74/4.0), confirming its effectiveness as an assistive healthcare technology. The integration of voice notifications specifically designed for elderly users with visual or cognitive impairments represents a significant advancement over traditional text-based reminder systems. The dependent monitoring fe ature effectively bridges communication gaps between elderly individuals and their caregivers, enabling proactive intervention when medication schedules are missed. The GIS mapping functionality enhances healthcare accessibility by providing real-time navigation to essential medical resources, addressing both routine medication management and emergency situations. Recommendations for future research include expanding the geographic coverage of the mapping database to include rural areas, implementing artificial intelligence for personalized medication adherence predictions, integrating with national health information systems, and conducting longitudinal studies to assess long-term impact on health outcomes. The system should incorporate machine learning algorithms to adapt to individual user patterns and provide increasingly personalized support. This research contributes to mobile health (mHealth) literature by demonstrating the practical application of voice-based assistive technologies combined with geographic information systems for elderly care in developing countries. MED-ALALAY provides a scalable, accessible solution that can be adapted for broader implementation across the Philippines and similar contexts, supporting healthy aging and reducing healthcare costs associated with medication non-adherence.

RECOMMENDATION

Based on the findings of the study and identified limitations, future researchers are encouraged to consider the proposed recommended points for better enhancement:

The system suggests that it needs to enhance the accuracy of locating places within the map. To ensure that all users who live in rural or urban-rural environments have access to essential healthcare services, creating a database of validated barangay health facilities, pharmacies, and clinics in the project area can be advantageous for the application to use 3D maps to enhance the user experience and locate broad local facilities.

It is recommended to enhance the voice notification feature by allowing users to customize the tone and add language for reminders to be more understandable, accessible, and engaging for elderly users who have hearing difficulties or language preferences. Using more comprehensive tools, applications, and software can be beneficial for elderly users to have greater clarity in the names of medicines when voice notification is enabled.

Future developers should conduct usability testing with a larger population of elderly participants to obtain more accurate data. A regular assessment using ISO 25010 software quality standards is also recommended to ensure that the system consistently fulfills the proper standards of reliability, accessibility, and performance efficiency.

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APPENDIX F.

Curriculum Vitae



Mark Jomar E. Aguilar

+639169082503

hawkaguilar13@gmail.com

Brgy. Malinta Los Baños Laguna

Career Objective

A disciplined and detail-oriented graduate seeking an entry-level role in a professional environment where strong ethical standards, effective communication, analytical thinking, and problem-solving abilities can be utilized to support organizational objectives and ensure high-quality performance.

Personal Information

Age: 21

Sex: Male

Nationality: Filipino

Civil Status: Single

Languages: English, Filipino

Skills

- Communication (verbal & written)
- Attention to detail
- Time management
- Presentation skills
- Problem-solving
- Critical thinking
- Work Ethic
- Creativity

Educational Background

Laguna State Polytechnic University (LSPU)

Bachelor of Science in Information Technology

Web and Mobile Application Development

2022 – Present

Bernardo N Calara Integrated National High School

Technical Vocational Livelihood

2020-2022



Althea De Villa

+639502238447

altheacsalme21@gmail.com

Brgy. La Mesa Calamba City, Laguna

Career Objective

As a highly motivated and driven BSIT graduate specializing in Web and Mobile Application Development. I am eager to leverage my technical expertise and passion for innovation. My academic projects enhanced my skills in problem-solving, collaboration, and teamwork that I can use in a real-world setting. I am seeking for an entry-level developer position in a dynamic team where i can contribute, learn, and grow professionally.

Personal Information

Age: 21

Sex: Male

Nationality: Filipino

Civil Status: Single

Languages: English, Filipino

Skills

- Problem-Solving
- Analytical Thinking
- Adaptability
- Communication Skill
- Teamwork
- Time Management
- Collaboration Skill

Educational Background

Laguna State Polytechnic University (LSPU)

Bachelor of Science in Information Technology

Web and Mobile Application Development

2022 – Present

CITI GLOBAL COLLEGE Calamba Campus

2020-2022



Geisler Tomboc

+63976419540

geislertomboc10@gmail.com

Brgy. Timugan Los Baños Laguna

Career Objective

To obtain a position in the field of Information Technology where I can apply my knowledge as a BSIT student specializing in Web and Mobile Application Development. I am currently contributing to our capstone system project, working collaboratively with a team, and am committed to building efficient, user-centered applications while continuously improving my technical skills and professional competence.

Personal Information

Age: 21

Sex: Male

Nationality: Filipino

Civil Status: Single

Languages: English, Filipino

Skills

Programming Languages

- HTML
- CSS
- PHP
- Java
- JavaScript
- TypeScript
- React js
- React Native

Front-End Frameworks/Librarie

- React.js
- Tailwind CS
- Bootstrap

Educational Background

Laguna State Polytechnic University (LSPU)

Bachelor of Science in Information Technology

Web and Mobile Application Development

2022 – Present

Christian Bible Baptist Academy Los Baños

General Academic Strand

2020-2022

APPENDIX G.

Turnitin & Grammarly Results

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